

Lecture 8: Morphological Image Processing

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Previously...

- Degradation model

- Noise only $g(x, y) = f(x, y) + \eta(x, y)$

- Noise Models: Gaussian, Rayleigh,
- Remove noise: Spatial Filtering
- Remove noise: Frequency Domain Filtering

- Noise and degradation

- Filter Estimation $g(x, y) = \mathcal{H}[f(x, y)] + \eta(x, y)$
- Wiener Filter, etc.

Outline

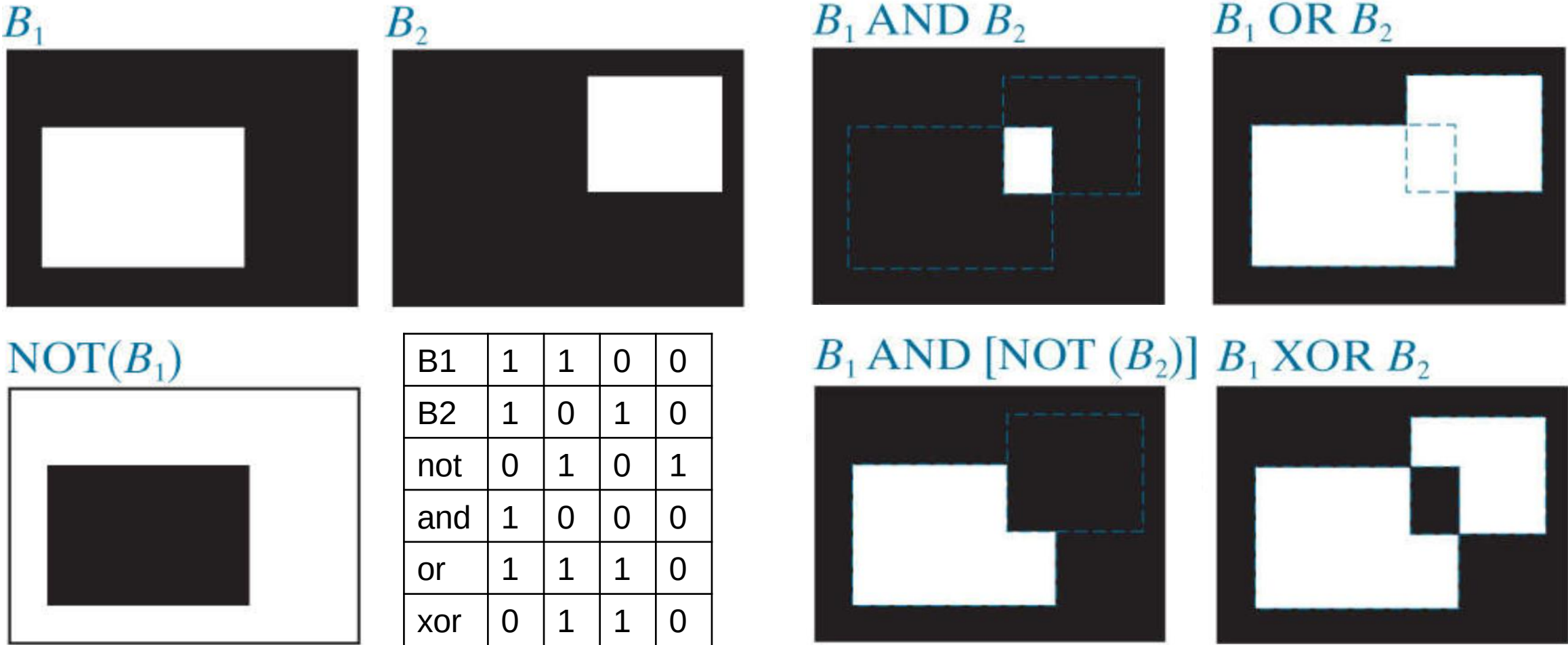
- Binary image
 - Preliminaries
 - Erosion and Dilation
 - Open and Close
 - Hit and miss
 - Basic algorithm (boundary detection, skeleton, ...)
- Grey scale image

Morphological operation

- Analysis of images in terms of shape (mostly binary)
- Advantages over other techniques:
 - Preserves edge information
 - Works by using shape-based processing
 - Can be designed to be idempotent
 - Computationally efficient

Preliminaries

- Logical operations



Preliminaries

Let A be a set in Z^2 . If $a = (a_1, a_2)$ is an element of A , then we write

$$a \in A$$

Similarly, if a is not an element of A , we write

$$a \notin A$$

An empty or *null* set is denoted by the symbol $\emptyset$.

$$A \subseteq B, \quad A \text{ is a subset of } B$$

$$C = A \cup B, \quad C \text{ is the } \textit{union} \text{ of } A \text{ and } B$$

$$D = A \cap B, \quad D \text{ is the } \textit{intersection} \text{ of } A \text{ and } B$$

$$A \cap B = \emptyset, \quad A \text{ and } B \text{ are } \textit{disjoint}$$

Preliminaries

The *complement* of a set A is the set of elements not contained in A :

$$A^c = \{w \mid w \notin A\}$$

The difference of two sets is defined as

$$A - B = \{w \mid w \in A, w \notin B\} = A \cap B^c$$

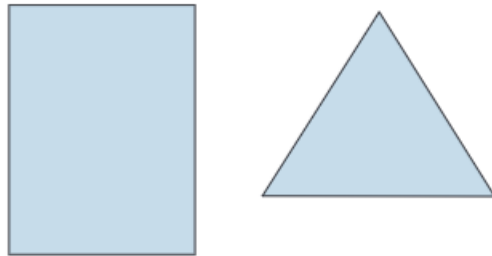
The *reflection* of a set B is defined as

$$\hat{B} = \{w \mid w = -b, \text{ for } b \in B\}$$

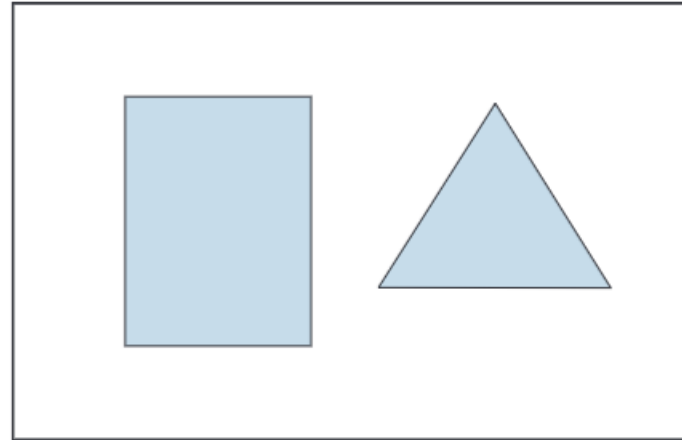
The *translation* of a set A by a point z is defined as

$$(A)_z = \{c \mid c = a + z, \text{ for } a \in A\}$$

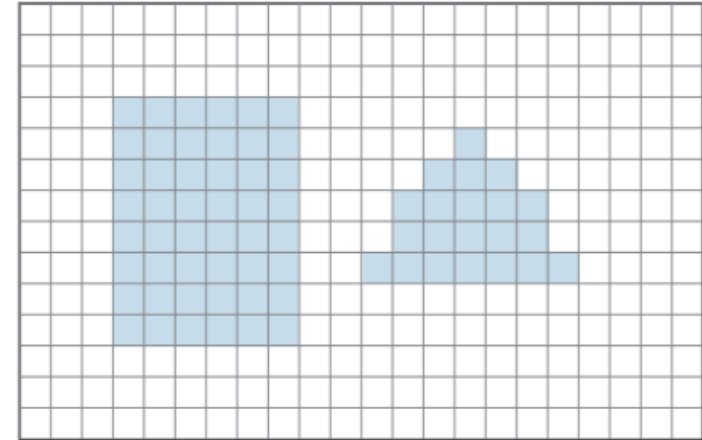
Structure Elements



Objects represented
as sets



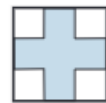
Objects represented as
a graphical image



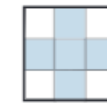
Digital image



Structuring element
represented as a set



Structuring element
represented as a graphical image



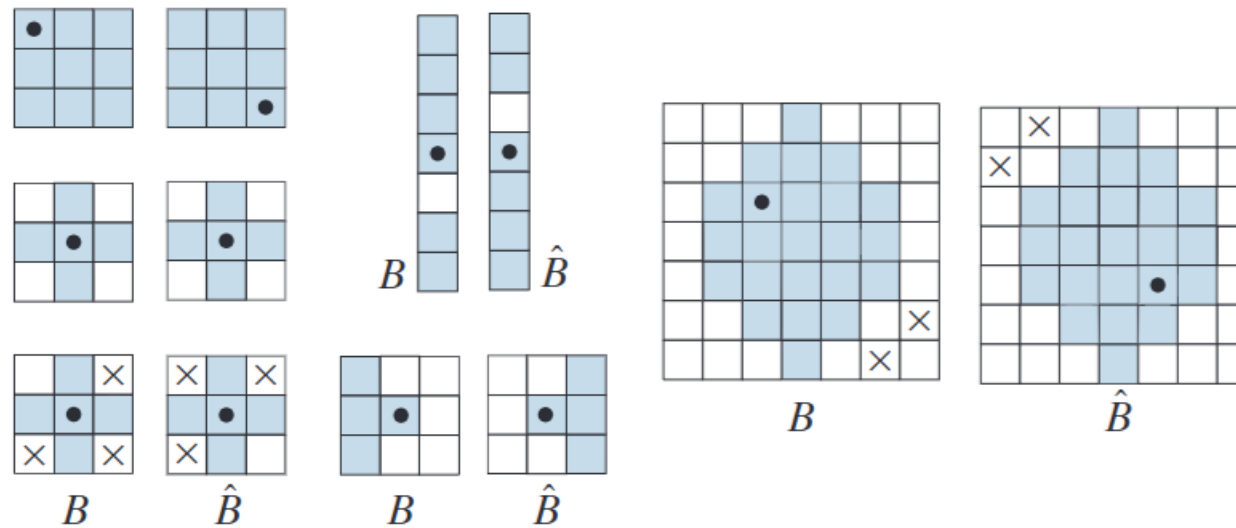
Digital
structuring element

Structure elements = convolution kernel

Structure Elements

Structure elements (SE) = convolution kernel

“x”: not care



The Erosion

- The erosion of A by B , denoted $A \ominus B$, is defined as

$$A \ominus B = \{z | (B)_z \in A\}$$

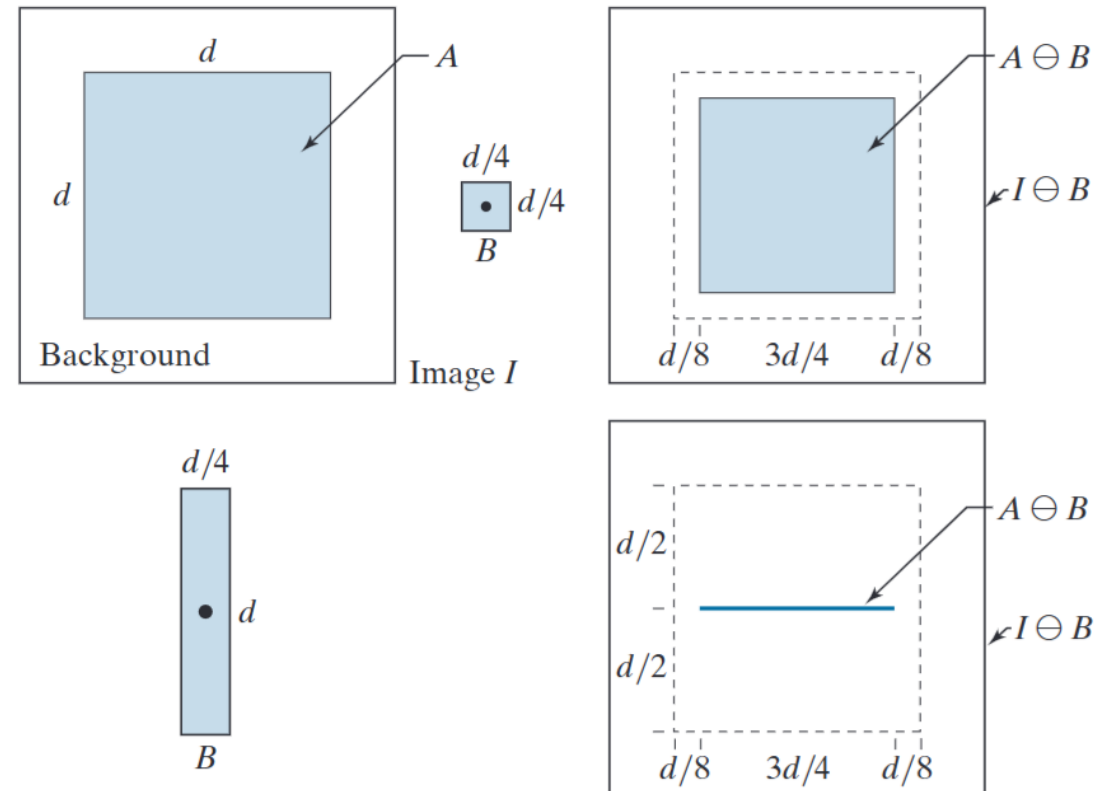
Set B is commonly referred to as the *structure element* here, and can be viewed as a convolution mask by this definition.

In general, the structure element has an analytically defined geometry (i.e., shape and size).

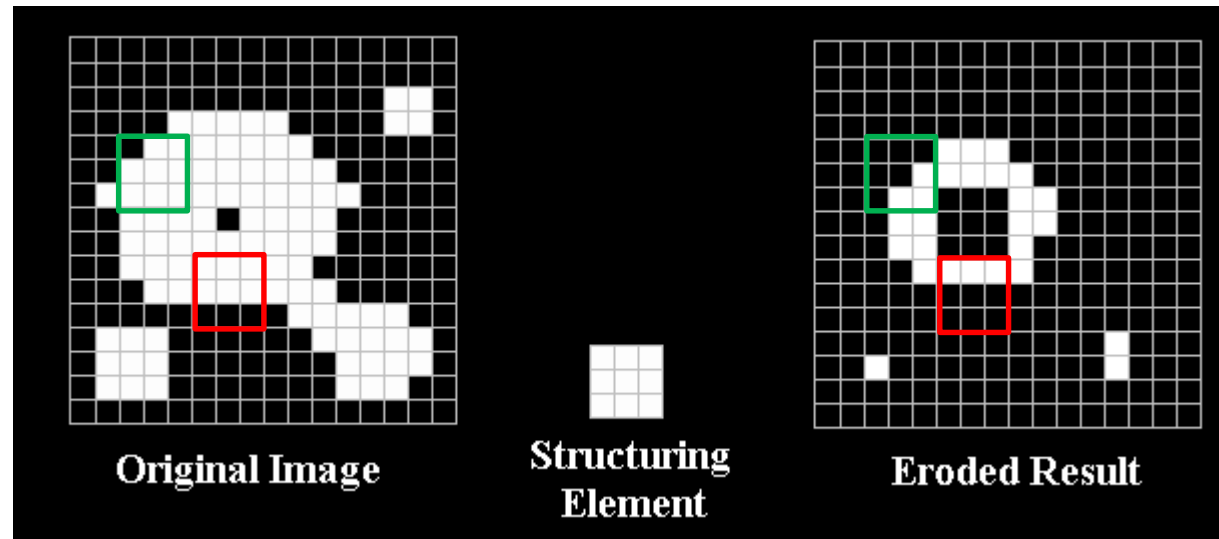
- This results in the set of all points z such that B , translated by z , is contained in A .

The Erosion

- Erosion is often thought of as shrinking or reducing operation. The reducing is achieved by peeling “layers” off an object.
- The structuring element is applied to the image by translating it across the image with its center origin point on every point of the image.

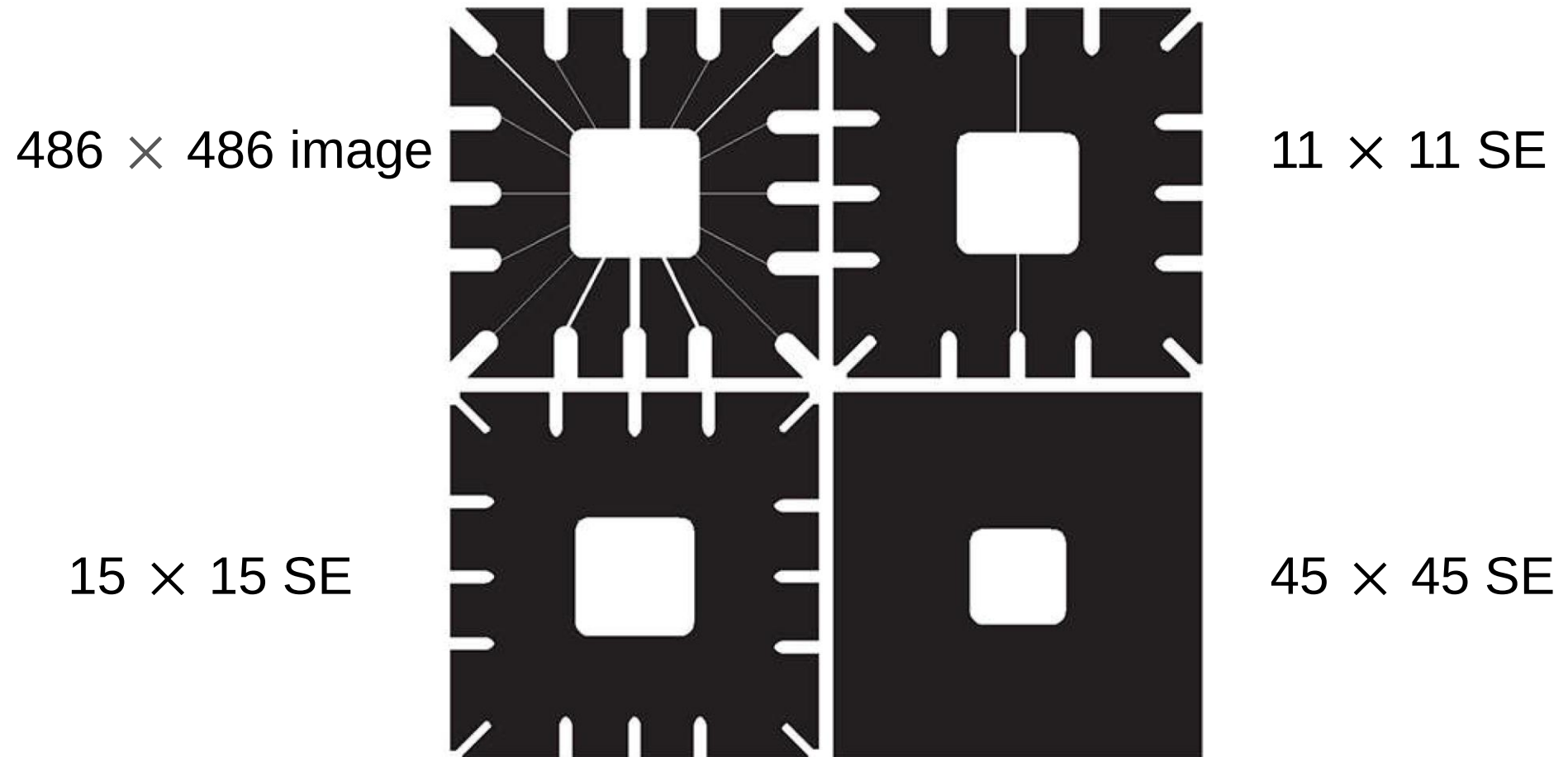


The Erosion



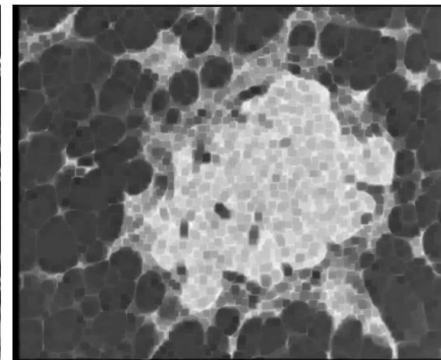
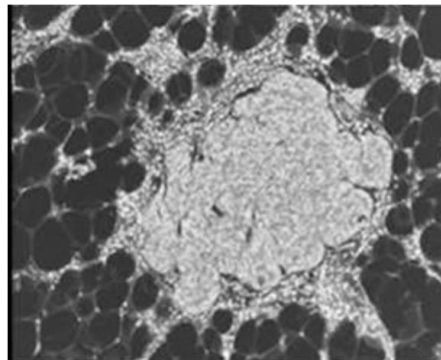
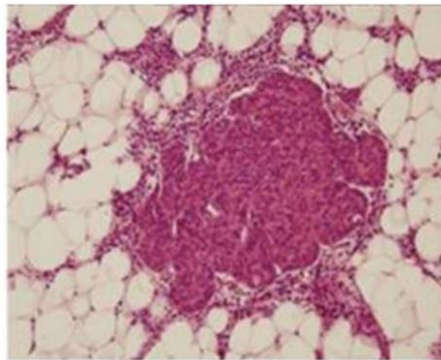
- Small objects are removed, layers are peeled off of larger objects, and connections between thinly connect objects are broken. Holes in an object are enlarged.
- One of the simplest applications of erosion is for eliminating irrelevant detail (in terms of size) from a binary image.

The Erosion

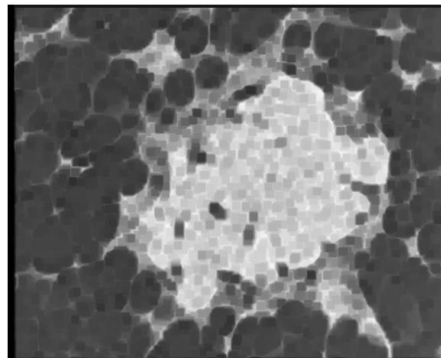


The Erosion

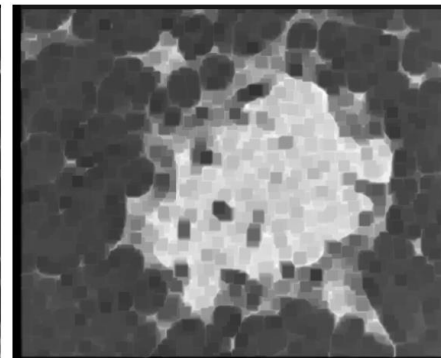
The thymic tissue component



7x7



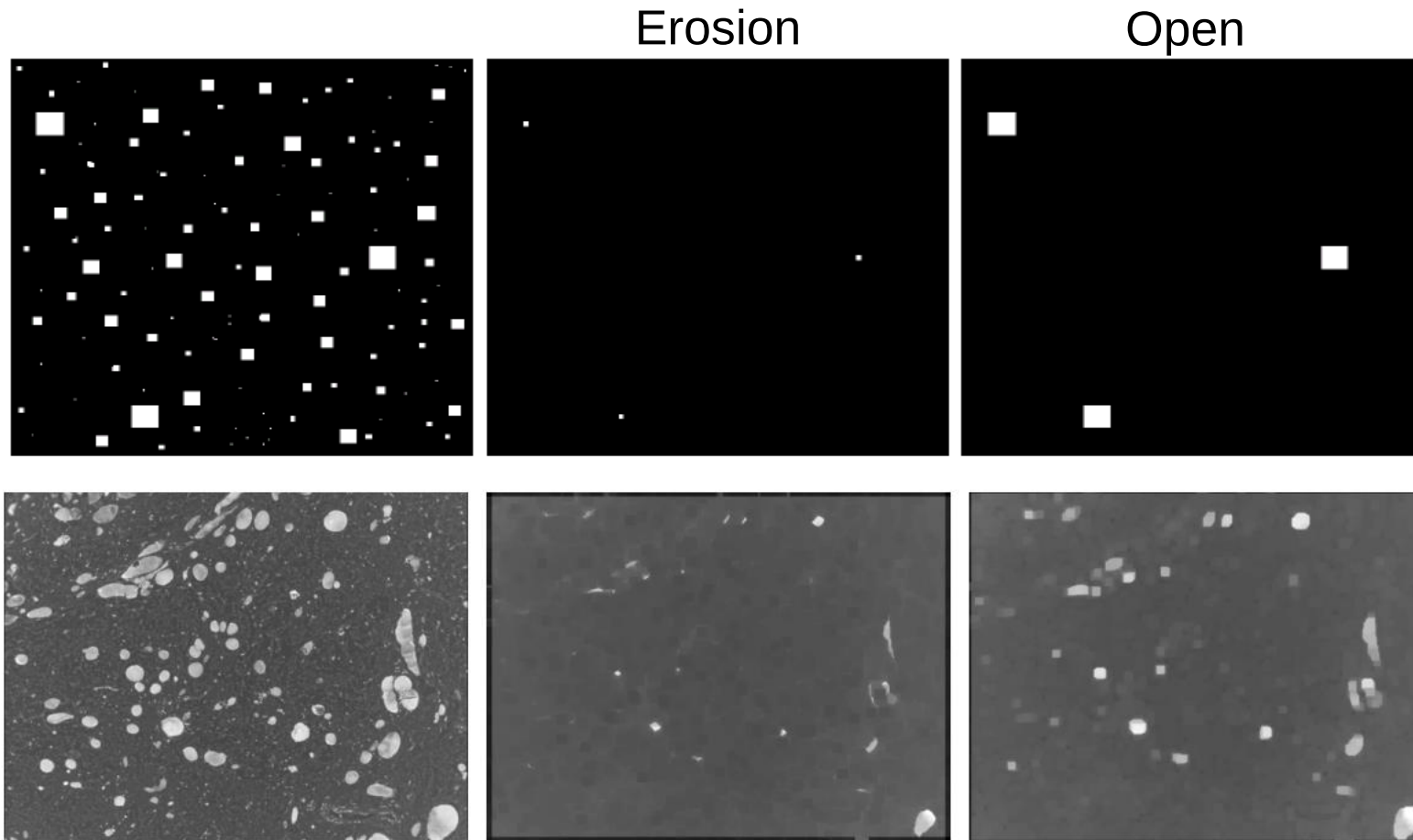
9x9



13x13

The Erosion

- Detect large object while removing small ones



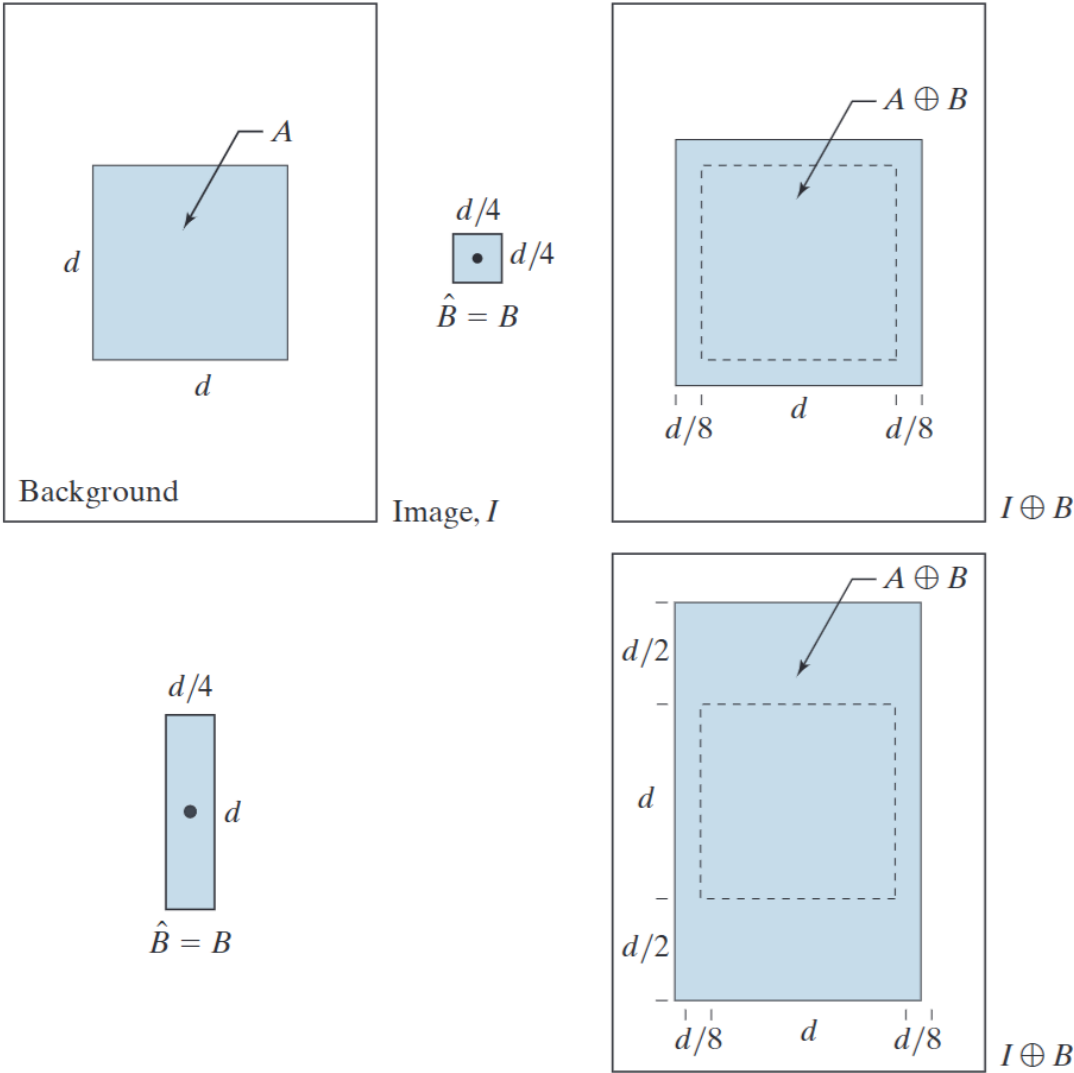
The Dilation

The *dilation* of A by B , denoted $A \oplus B$, is defined as

$$A \oplus B = \{z | (\hat{B})_z \cap A \neq \emptyset\}$$

- This results in the set of all points z such that B , translated by z , still overlaps with A .

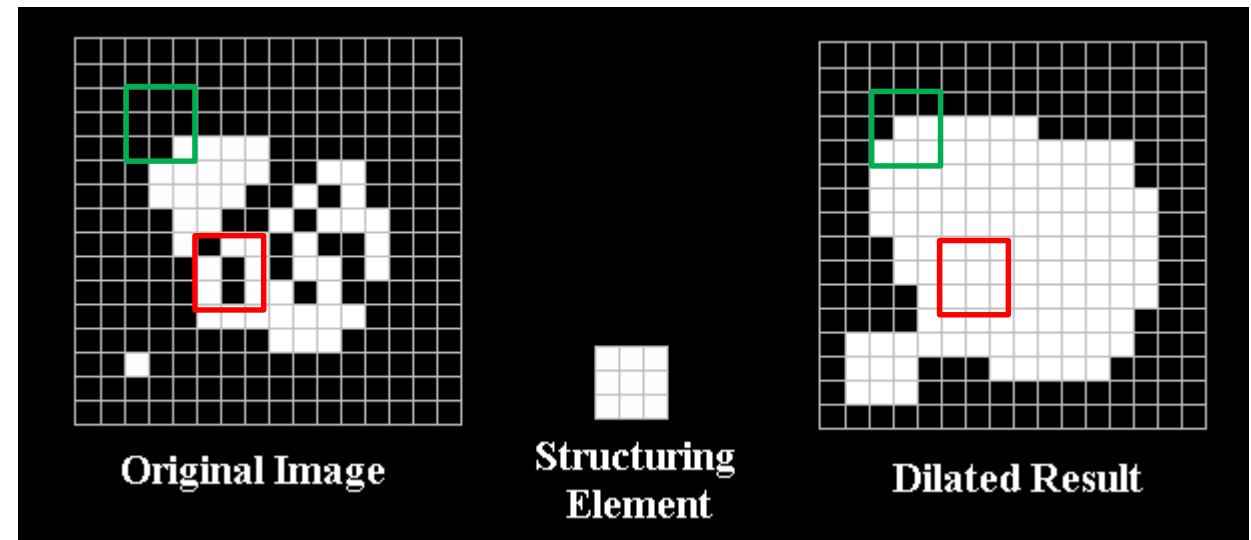
The Dilation



The Dilation

$$A \oplus B = \{z | (B)_z \cap A \neq \emptyset\}$$

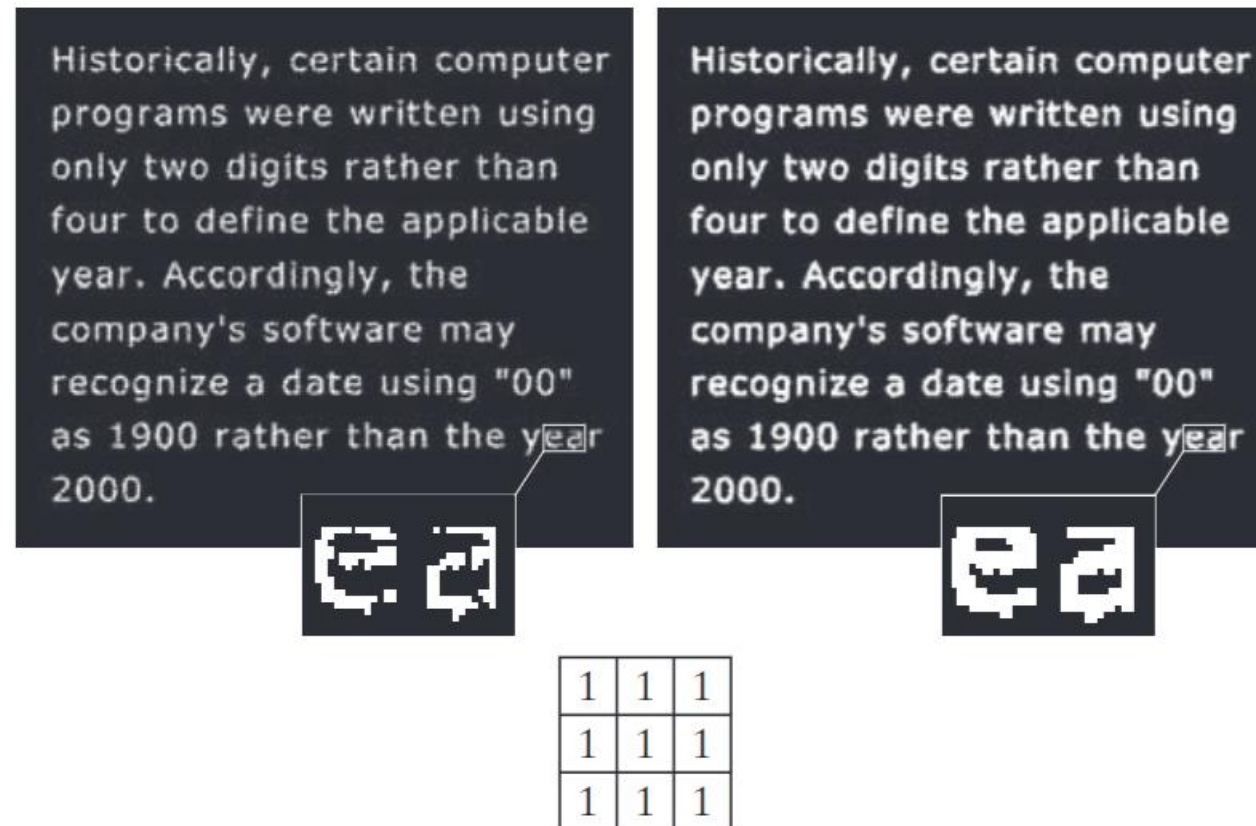
- Dilation is swelling or enlarging operation. The enlarging is achieved by adding “layers” on an object.
- Holes and cracks in an object are filled in, and layers are added to objects.



A Dilation Example

One of the simplest applications of dilation is for bridging gaps.

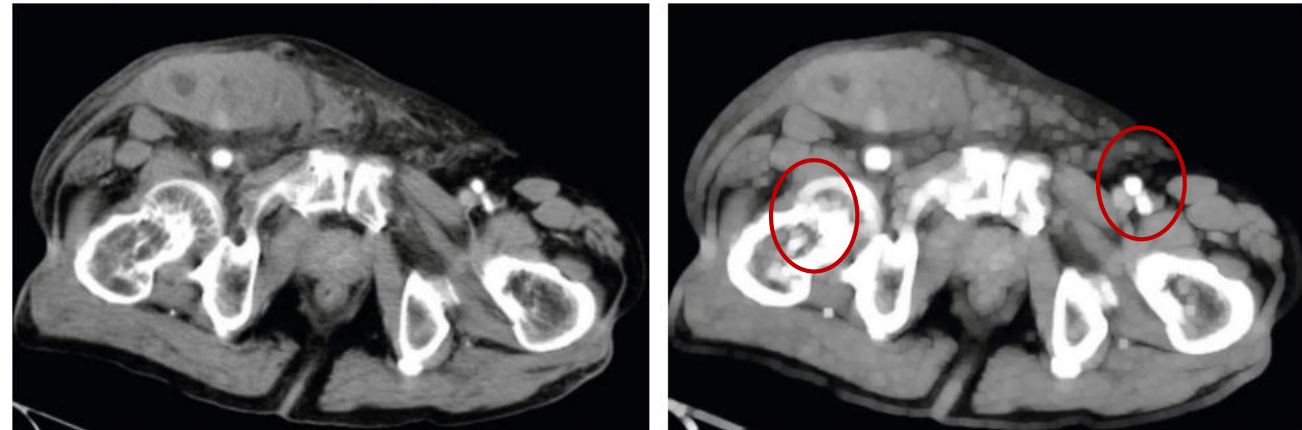
- The maximum length of the breaks is known to be two pixels



A Dilation Example

One of the simplest applications of dilation is for bridging gaps.

- The maximum length of the breaks is known to be two pixels



1	1	1
1	1	1
1	1	1

Some Properties of Dilation and Erosion

Translation invariant:

$$(A \oplus B)_z = A_z \oplus B$$

Order invariant:

$$(A \oplus B) \oplus C = A \oplus (B \oplus C)$$

Increasing operator:

$$A \subseteq B \Rightarrow A \oplus C \subseteq B \oplus C$$

Scale invariant:

$$rA \oplus rB = r(A \oplus B), \text{ } r \text{ is a scale factor}$$

Same with erosion.

The Relationship of Dilation and Erosion

- Dilation and erosion are duals of each other with respect to set complementation and reflection. That is,

$$(A \ominus B)^c = A^c \oplus \hat{B} \iff (A \oplus B)^c = A^c \ominus \hat{B}$$

$$A \ominus B = \{z | (B)_z \in A\}$$

$$A \ominus B = \{z | (B)_z \cap A^c = \emptyset\}$$

$$(A \ominus B)^c = \{z | (B)_z \cap A^c \neq \emptyset\}$$

$$(C = \emptyset)^c \Rightarrow C \neq \emptyset$$

$$\begin{aligned} (A \ominus B)^c &= \{z | (B)_z \cap A^c \neq \emptyset\} \\ &= A^c \oplus \hat{B} \end{aligned}$$

Opening and Closing

- Erosion and dilation $\rightarrow$ opening and closing
- The opening of set A by structure element B is defined as

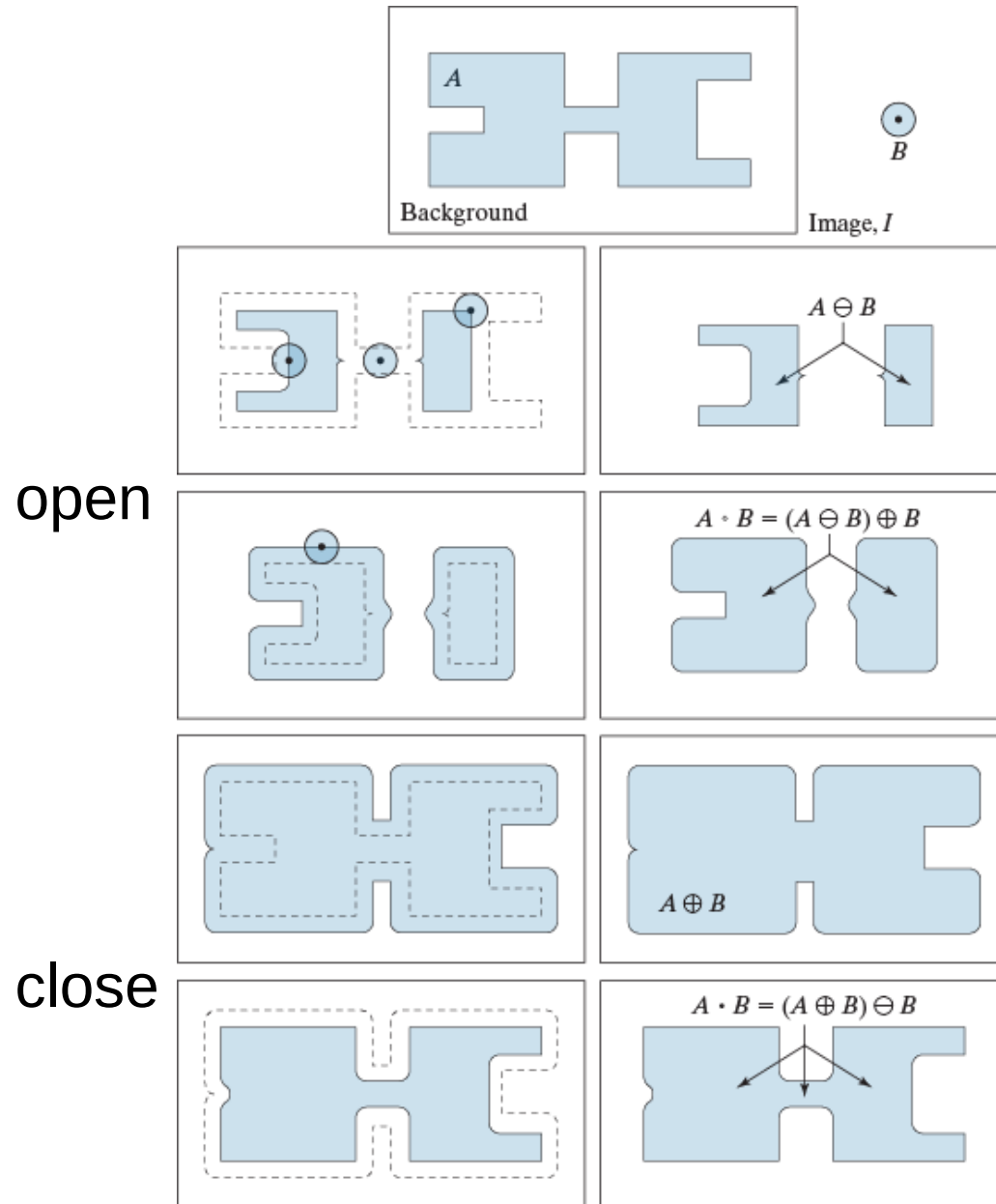
$$A \circ B = (A \ominus B) \oplus B$$

Opening generally smooths the contour of an object, breaks narrow isthmuses, and eliminates thin protrusions.

- The closing of set A by structure element B is defined as

$$A \cdot B = (A \oplus B) \ominus B$$

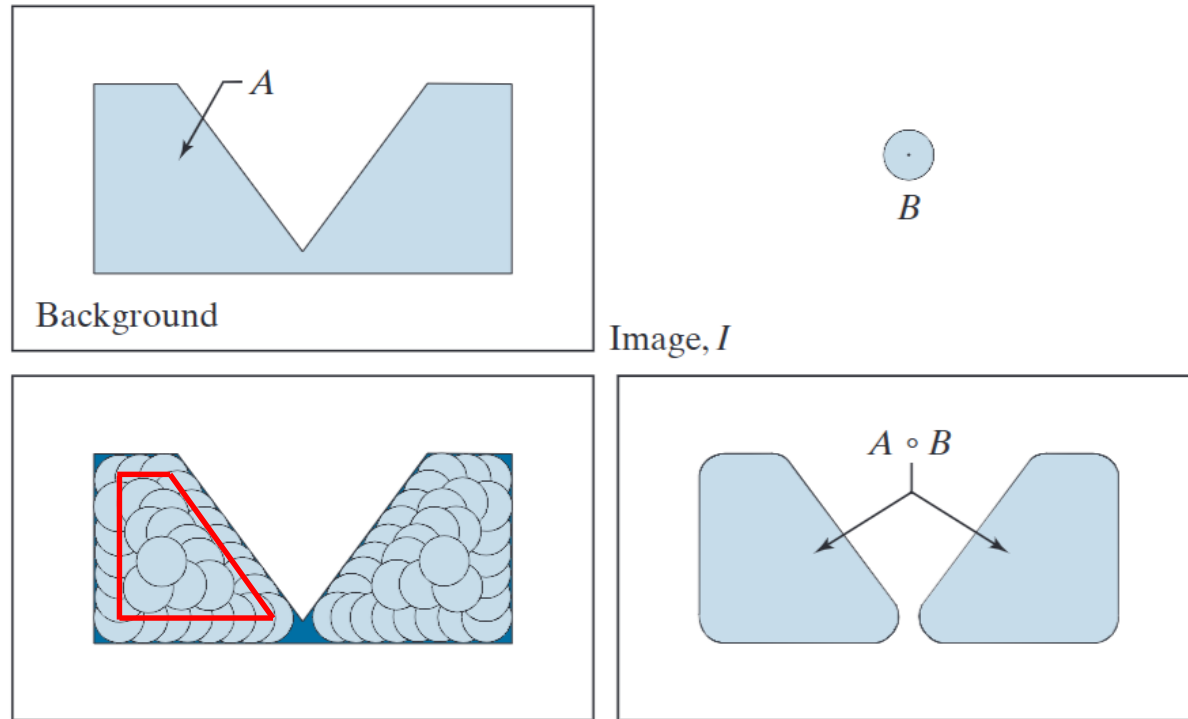
Closing also tends to smooth sections of contours, fuses narrow breaks and long thin gulfs, eliminates small holes, and fills gaps.



Opening and Closing Operations

- The bridge and protruding elements were eliminated.
- The outward pointing corners were rounded, whereas inward pointing corners were not affected.
- The inward pointing corners were rounded, whereas outward pointing corners were not affected.
- The leftmost intrusion on the boundary of A was reduced in size significantly.

Geometric Interpretation of Opening and Closing

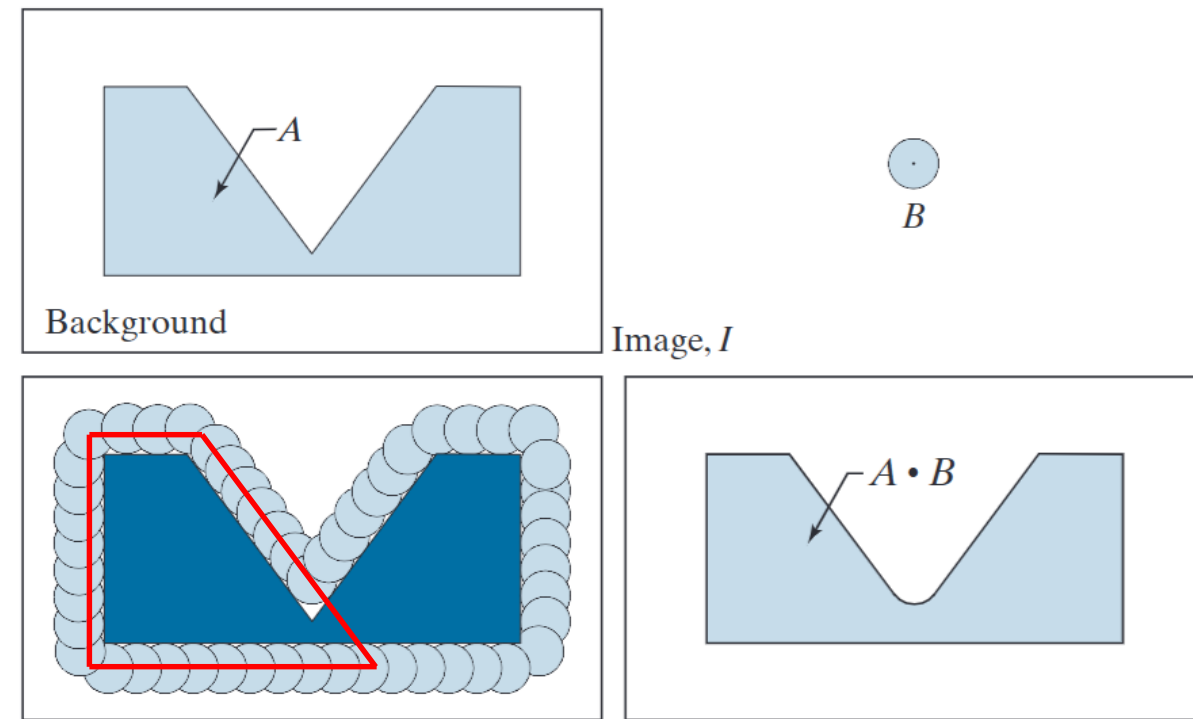


open

$$A \circ B = (A \ominus B) \oplus B$$

$$A \circ B = \cup \{(B_z) | B_z \subseteq A\}$$

Erosion: $A \ominus B = \{z | (B)_z \in A\}$



close

$$A \cdot B = (A \oplus B) \ominus B$$

$$A \cdot B = \cup \{(B_z) | B_z \cap A = \emptyset\}^c$$

Properties of Opening and Closing

Like dilation and erosion, opening and closing are duals of each other with respect to set complementation and reflection. That is,

$$(A \bullet B)^c = A^c \circ \hat{B}$$

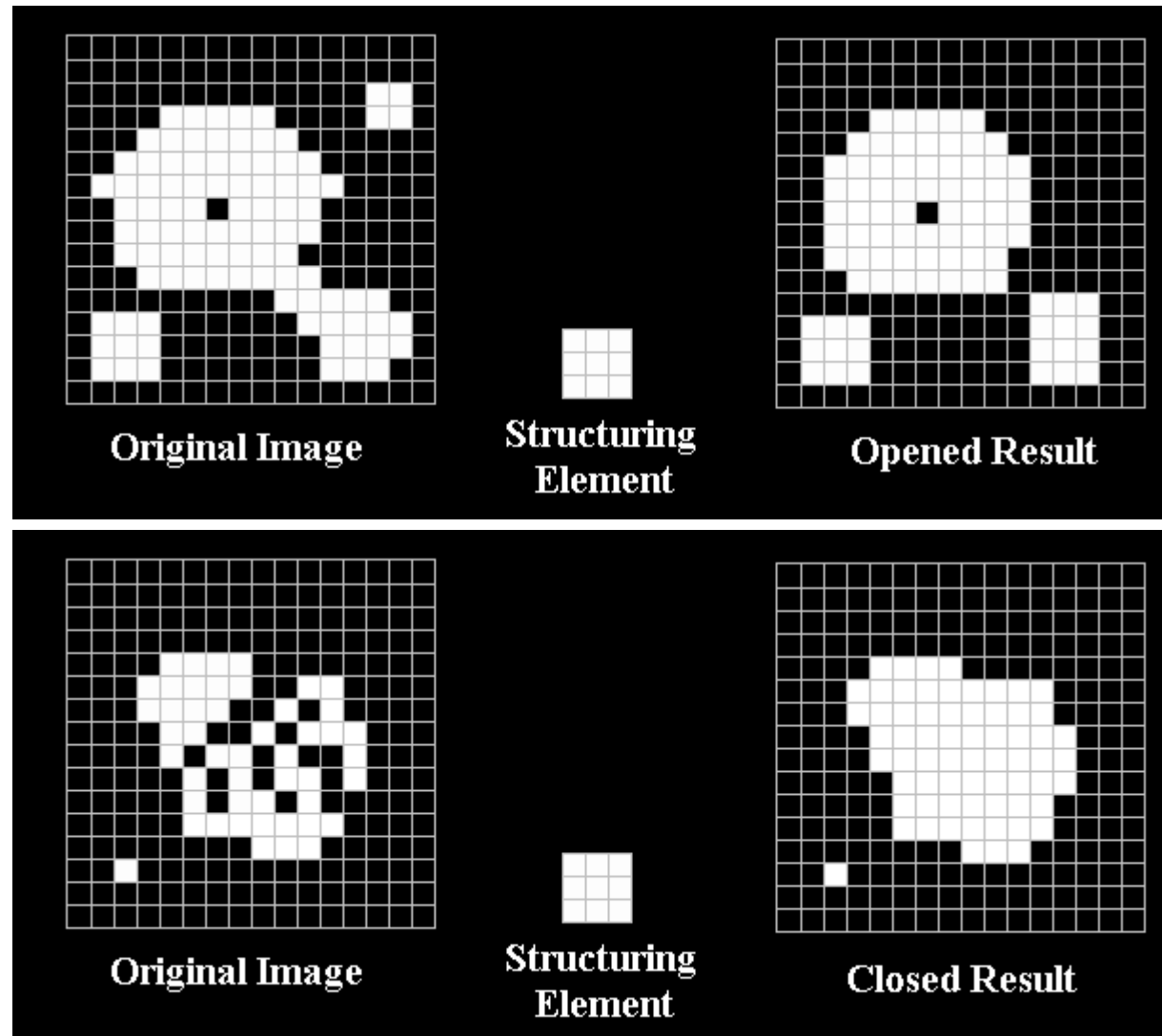
The opening operation satisfies the following properties:

- (i) $A \circ B$ is a subset (subimage) of A .
- (ii) If C is a subset of D , then $C \circ B$ is a subset of $D \circ B$.
- (iii) $(A \circ B) \circ B = A \circ B$ (idempotent)

Similarly, The closing operation satisfies the following properties:

- (i) A is a subset (subimage) of $A \bullet B$.
- (ii) If C is a subset of D , then $C \bullet B$ is a subset of $D \bullet B$.
- (iii) $(A \bullet B) \bullet B = A \bullet B$ (idempotent)

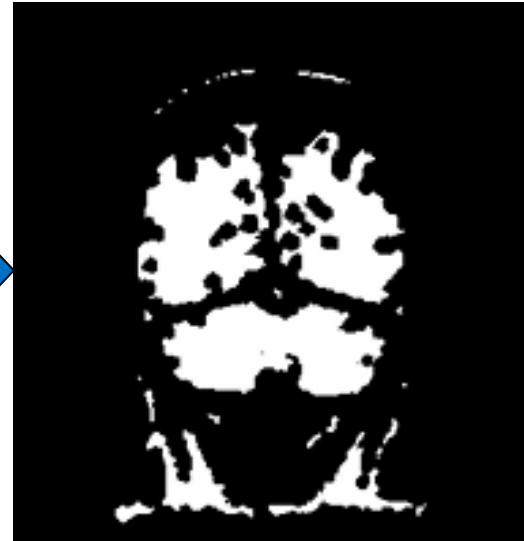
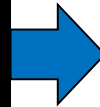
Opening and Closing Examples



An Example of Opening: Skull stripping

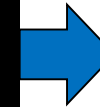


Binary brain image



Erosion

Circular structure
element of radius 3

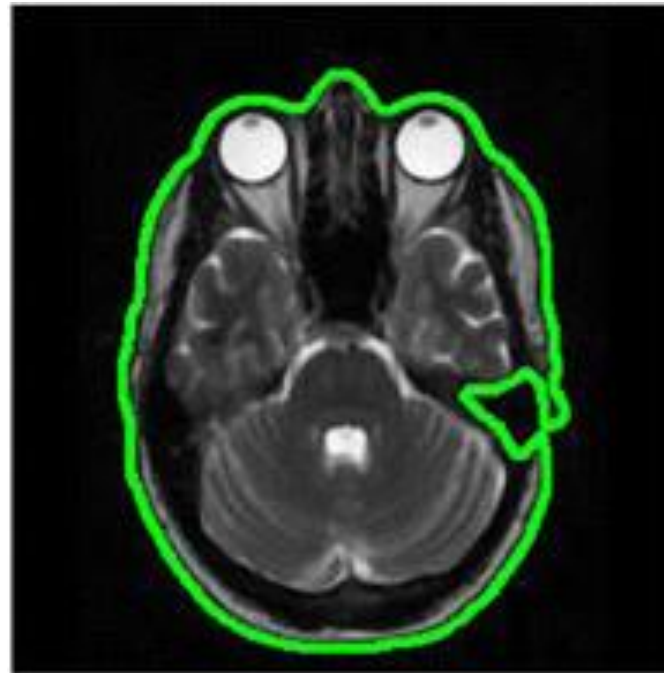


Conditional dilation

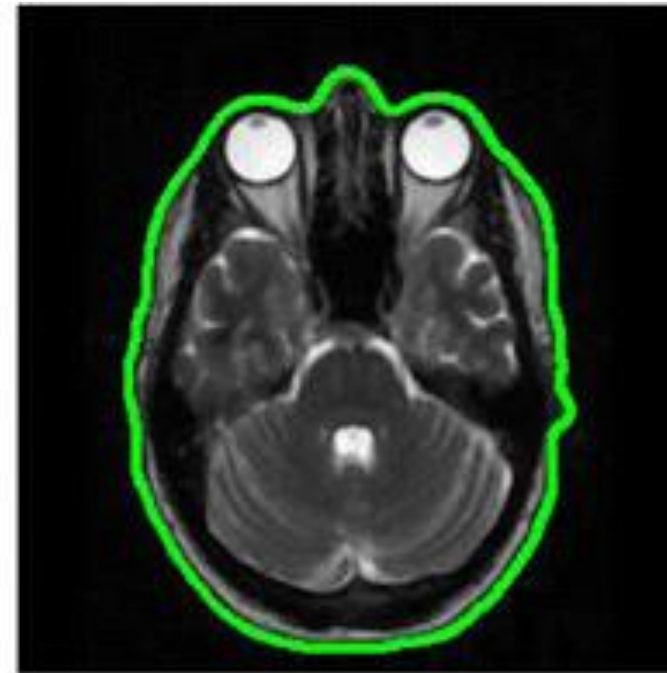
Circular structure
element of radius 4

Largest
connected
component

An Example of Closing: ROI modification



without closing



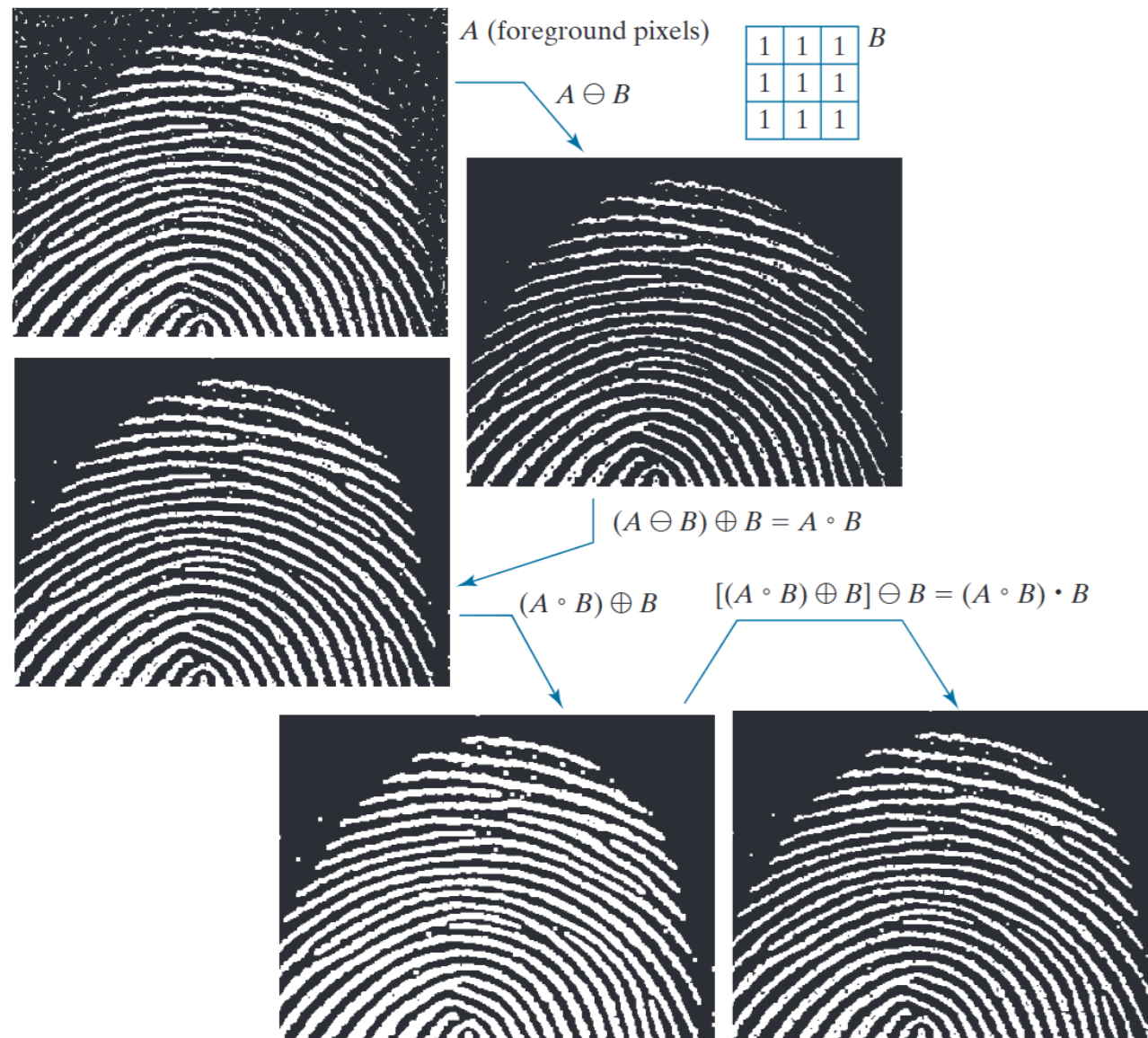
with closing

Morphological Filter – Opening then Closing

- Eliminate salt and pepper' noise
- The background noise is eliminated at the erosion stage
- All noise components are physically smaller than the structure element B .
- Preserve the main geometric structures of the object.
 - Only features smaller than the structure element are affected by transformations.
 - **All other features at larger scales are not degraded.** This is not the case with linear transformations, such as convolution



An Example of Opening Closing filter



The Hit-or-Miss Transformation

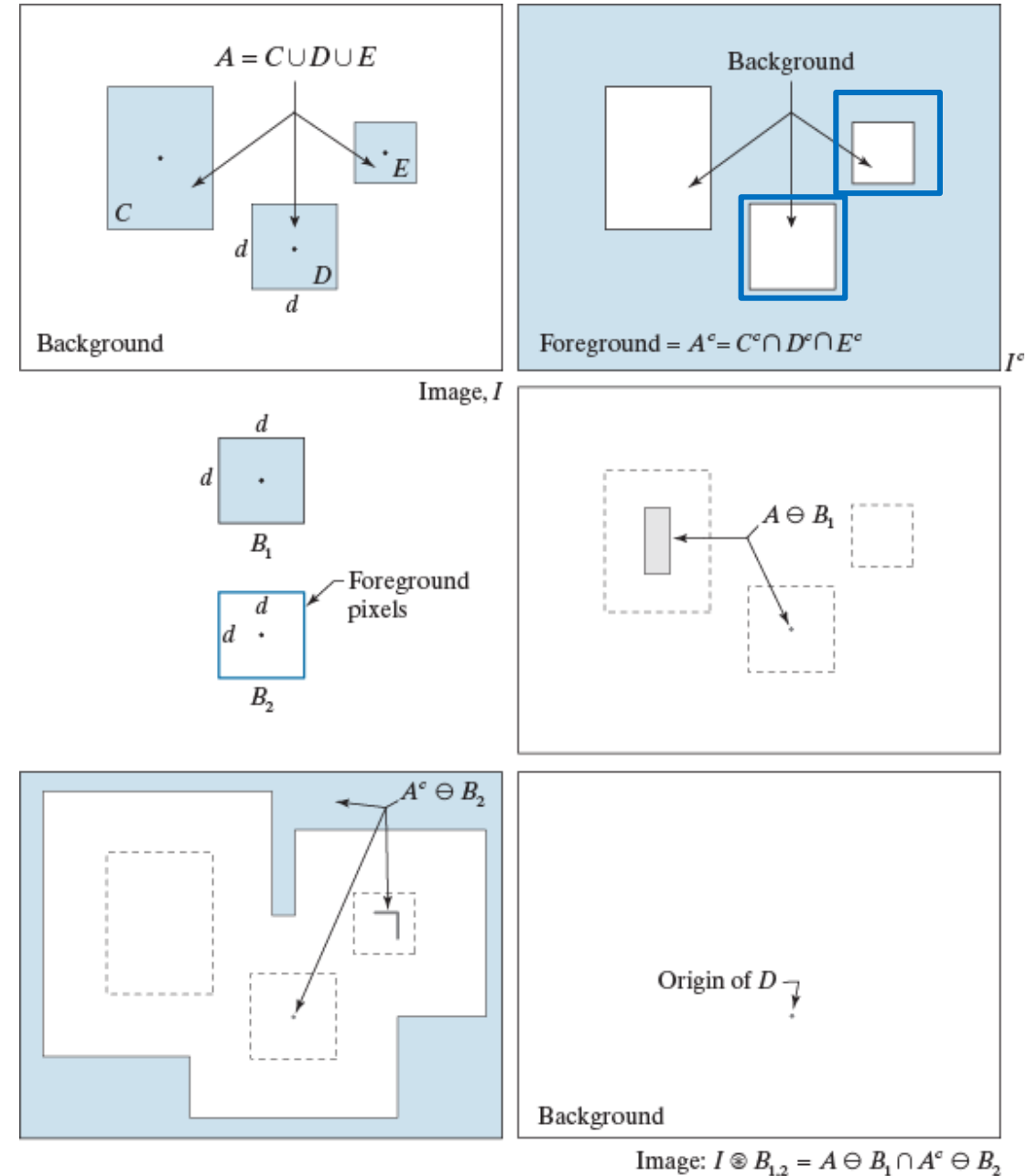
Shape detection

If B denotes the set composed of X and its background, the match of B in A , denoted as $A \circledast B$, is

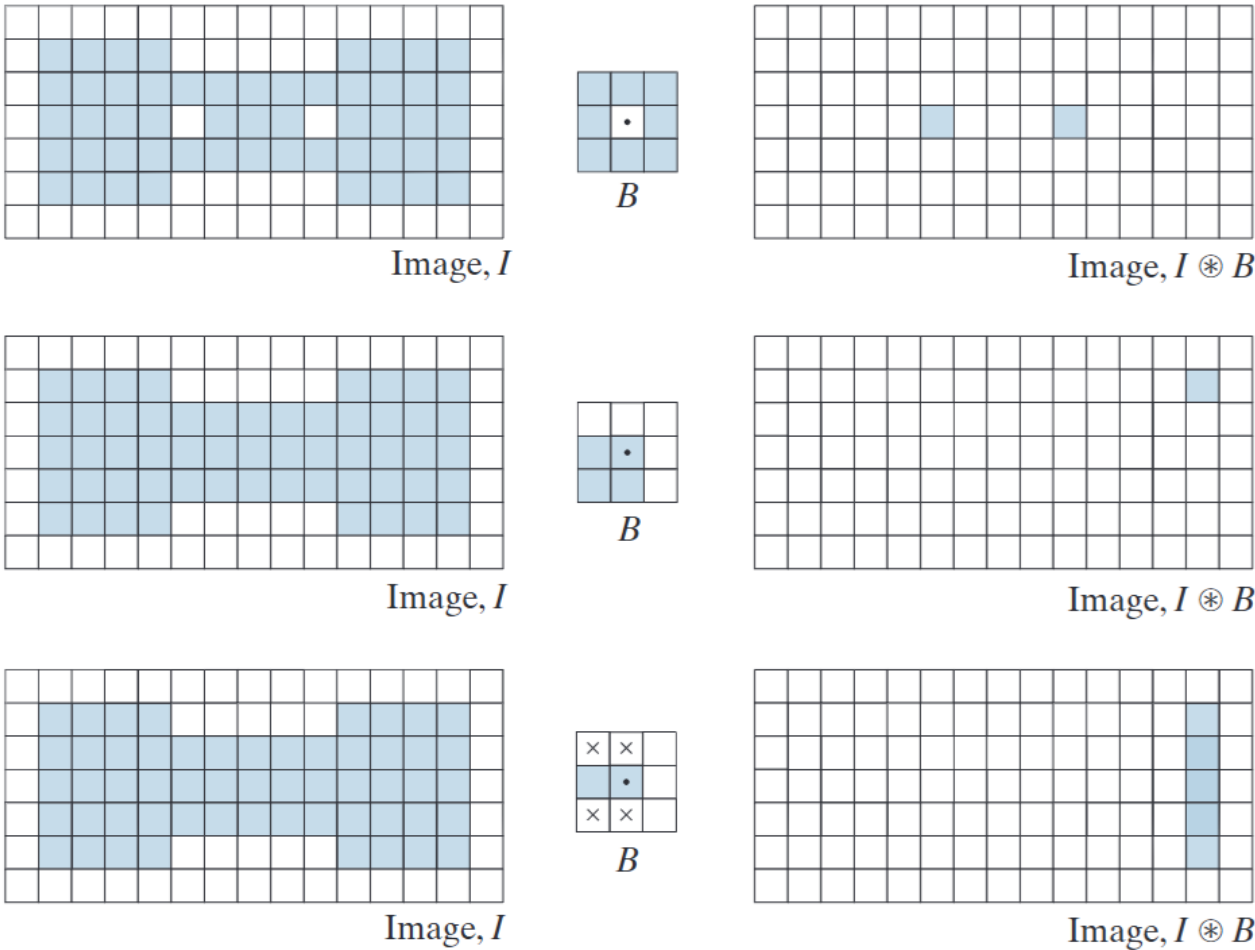
$$A \circledast B = (A \ominus B_1) \cap (A^c \ominus B_2)$$

B1		
1	1	1
1	1	1
1	1	1

B2				
1	1	1	1	1
1	0	0	0	1
1	0	0	0	1
1	0	0	0	1
1	1	1	1	1



The Hit-or-Miss Transformation



Find a match of B in I

Some Basic Morphological Algorithms

When dealing with binary images, the principal application of morphology is extracting image components that are useful in the representation and description of shape.

- ❖ boundaries
- ❖ connected components
- ❖ convex hull
- ❖ skeleton

We also develop several methods that are used frequently in conjunction with above algorithms as pre- or postprocessing steps.

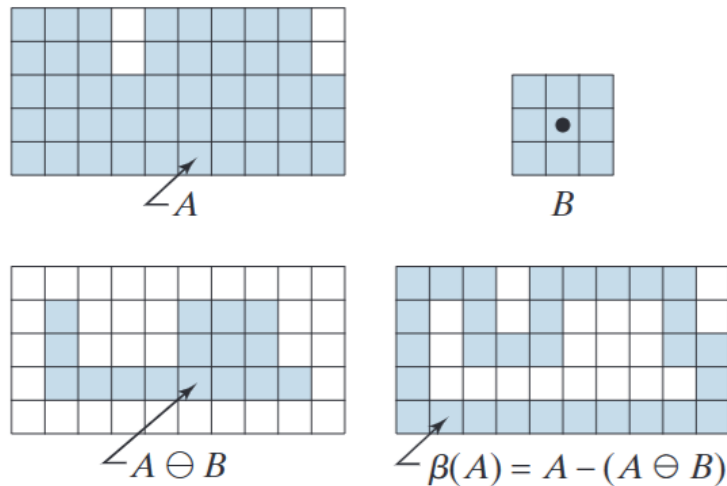
- ❖ region filling
- ❖ thinning
- ❖ thickening
- ❖ pruning

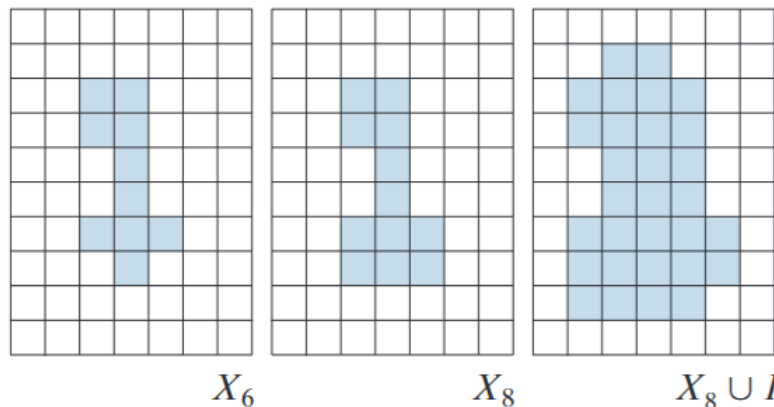
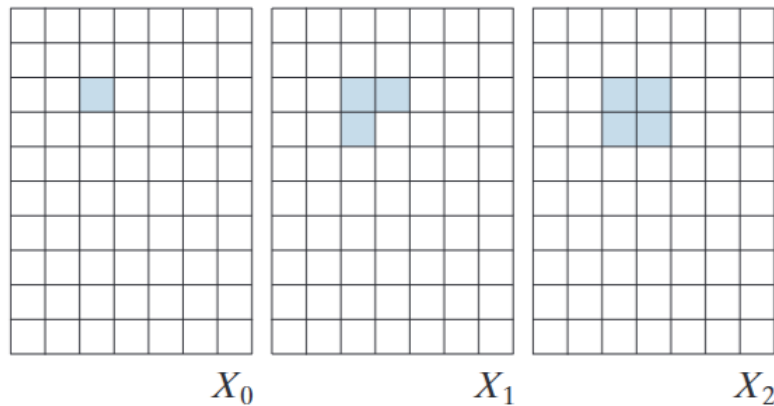
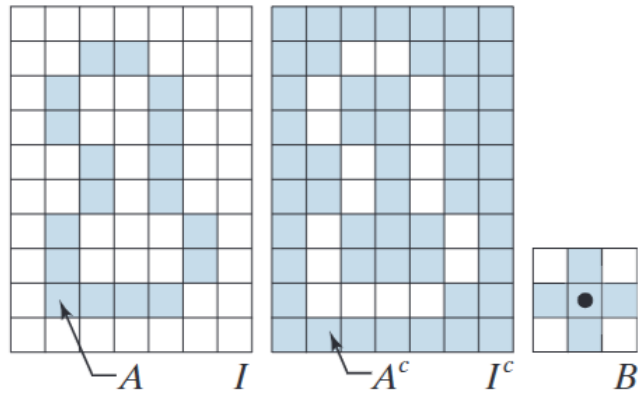
Boundary Extraction

The boundary of a set A denoted by $\beta(A)$, can be obtained by:

$$\beta(A) = A - (A \ominus B)$$

where B is a suitable structuring element.





Hole Filling

We know a start point p inside the boundary, the following procedure will fill the region with 1's:

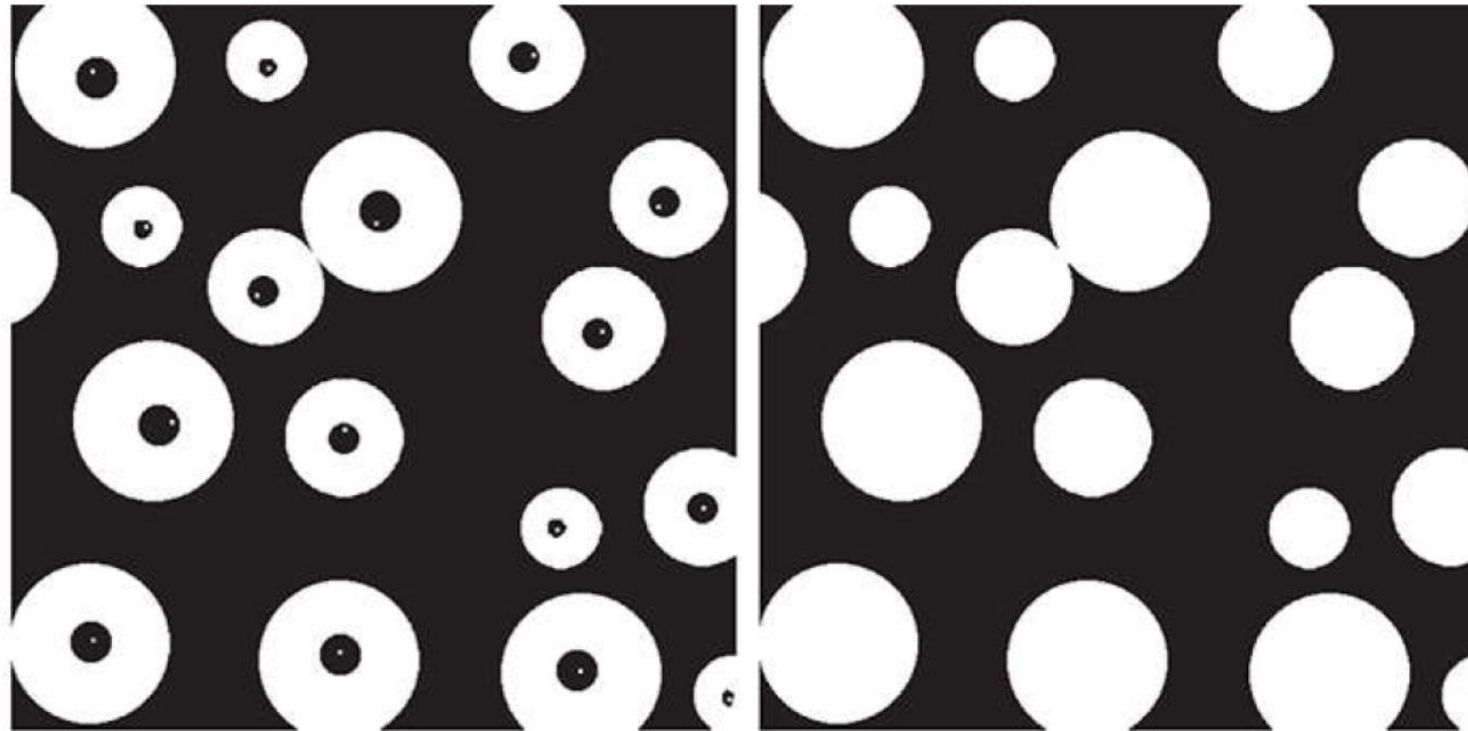
$$X_k = (X_{k-1} \oplus B) \cap A^c \quad k = 1, 2, 3, \dots$$

where we assign a value of 1 to p to begin, and $X_0 = p$. the algorithm terminates at iteration step k if $X_k = X_{k-1}$

$$X_8 = \text{hole}$$

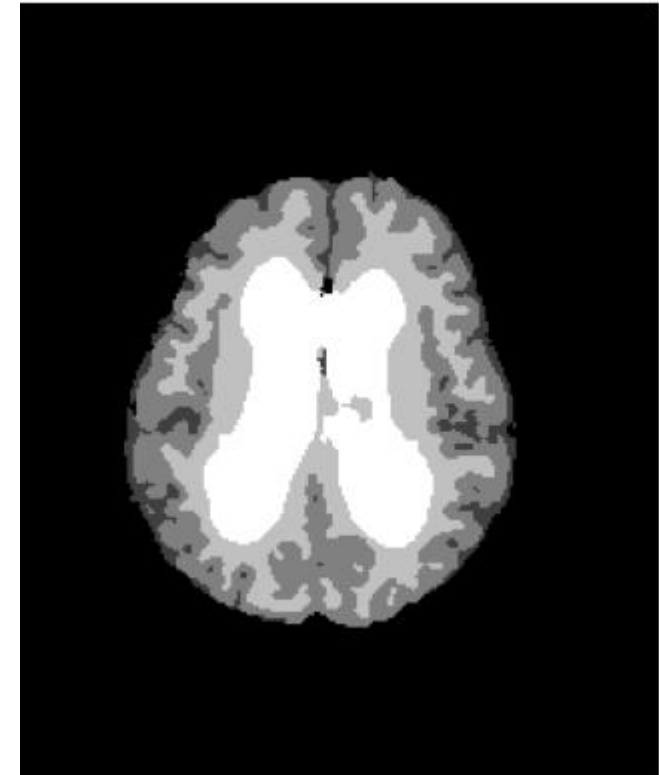
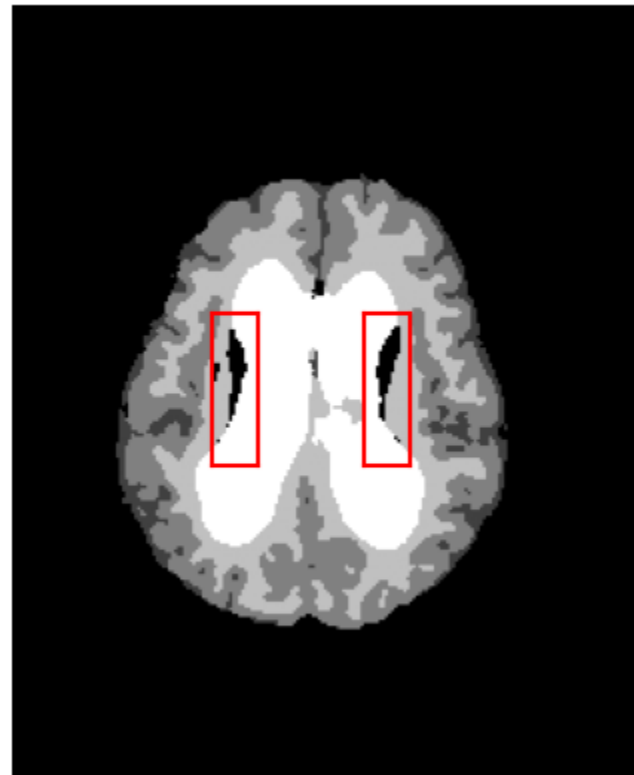
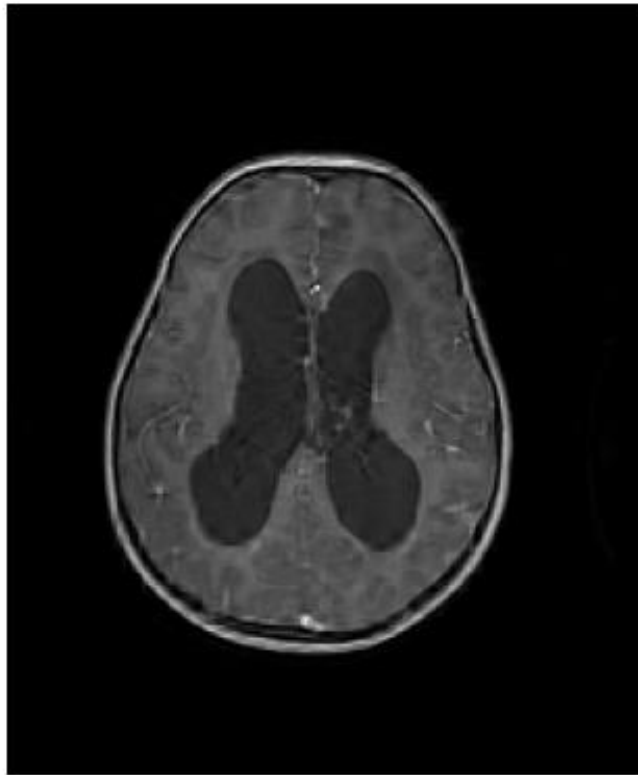
Intersection at each step with limits the result to inside the region of interest.

An Example



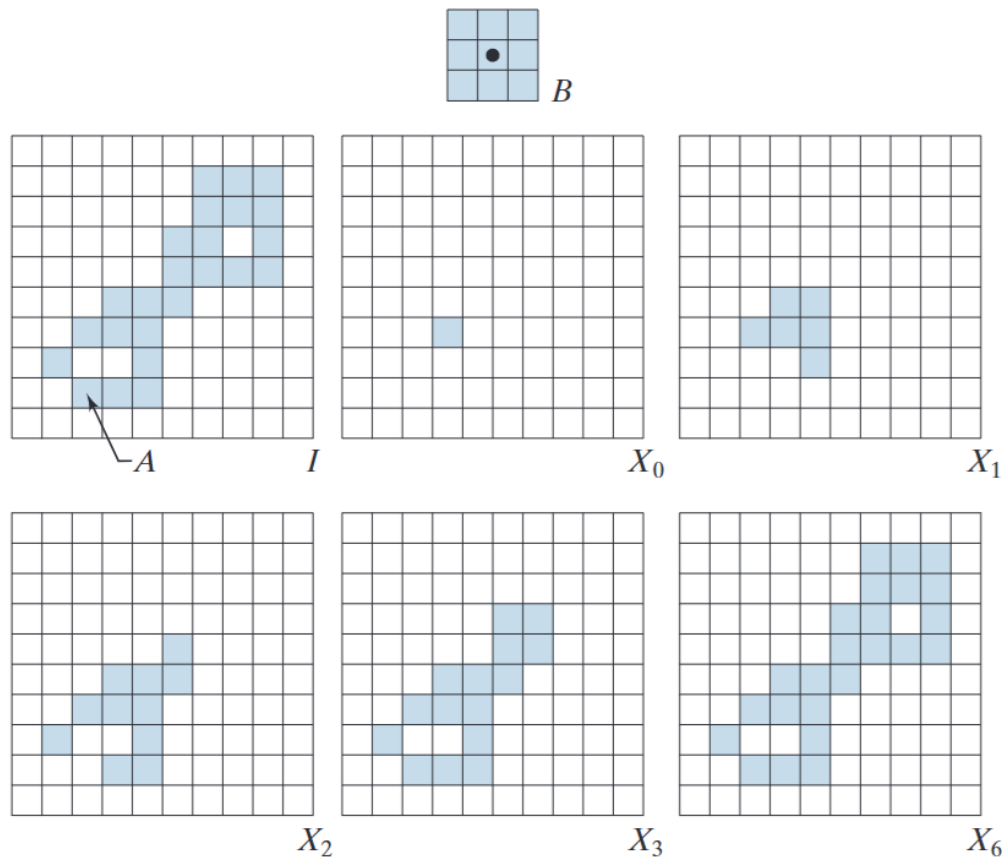
An Example

Brain segmentation mask



Extracting of Connected Components

Let Y represents a connected components contained in a set A



$$X_k = (X_{k-1} \oplus B) \cap A \quad k = 1, 2, 3, \dots$$

where $X_0 = p$ (p inside each component)

The algorithm converges when $X_k = X_{k-1}$ and we let $Y = X_k$

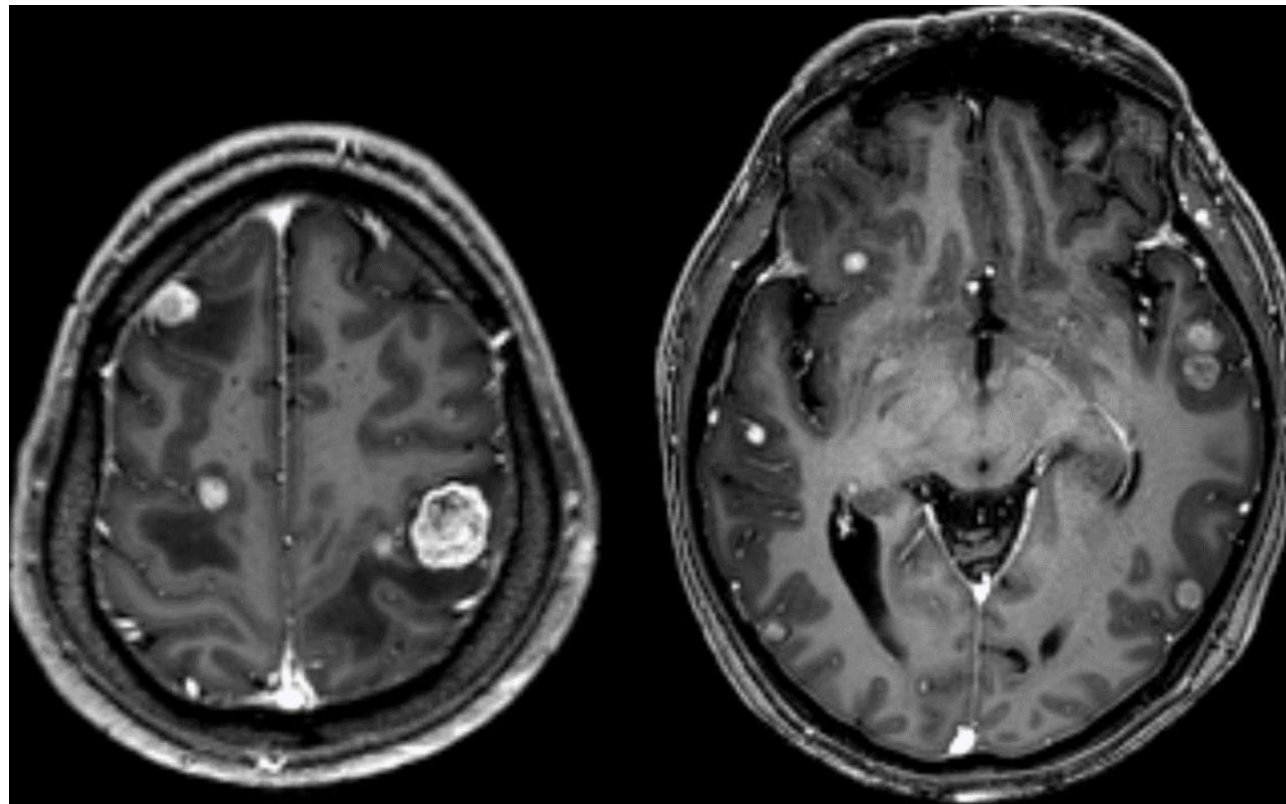
An Example



Connected component	No. of pixels in connected comp
01	11
02	9
03	9
04	39
05	133
06	1
07	1
08	743
09	7
10	11
11	11
12	9
13	9
14	674
15	85

An Example

Brain metastasis



Thinning

The thinning of a set A by a structure element B can be defined by:

$$\begin{aligned} A \otimes B &= A - (A \circledast B) \\ &= A \cap (A \circledast B)^c \end{aligned}$$

A more useful expression for thinning A symmetrically is based on a sequence of structure elements:

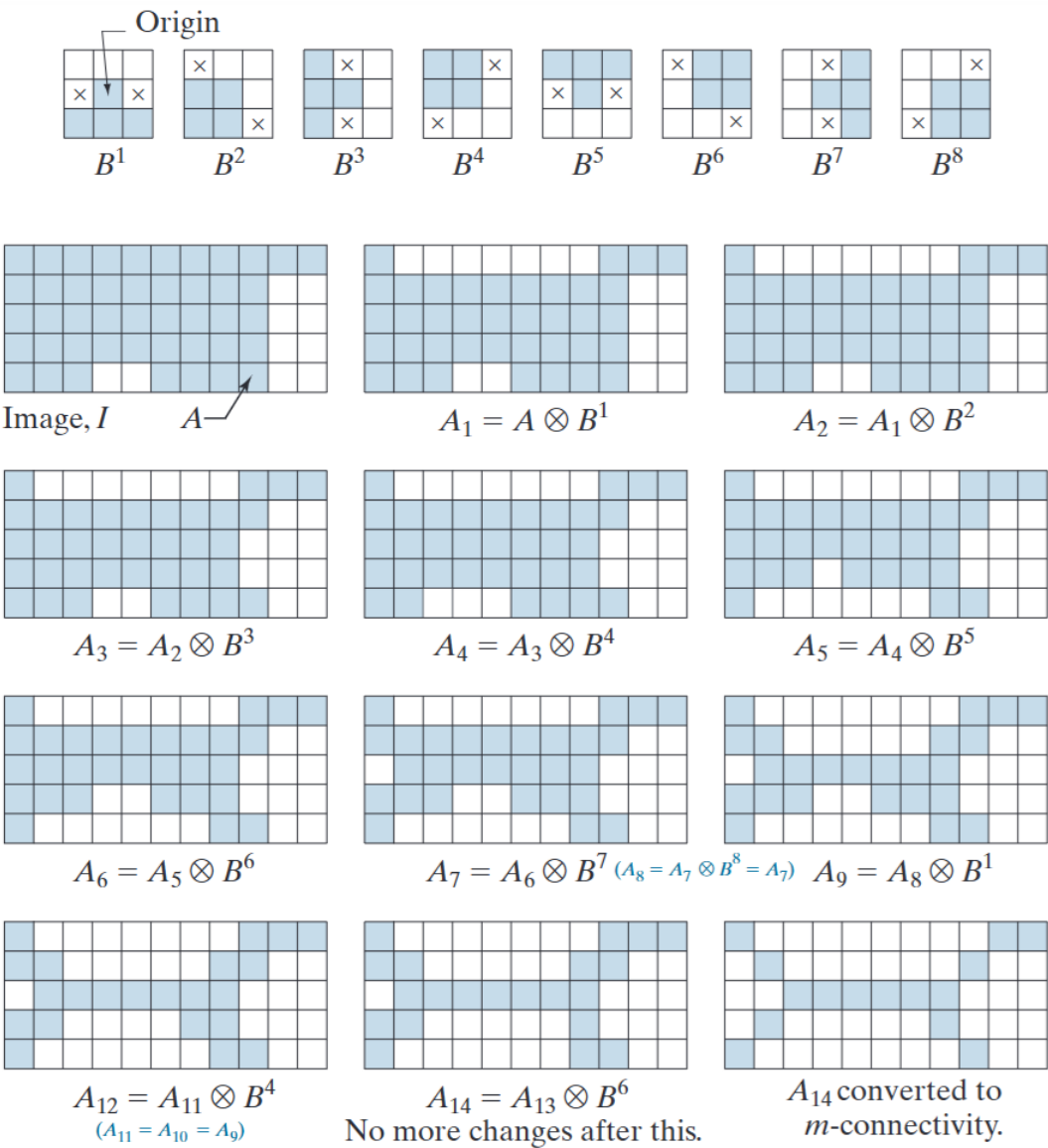
$$\{B\} = \{B^1, B^2, B^3, \dots, B^n\}$$

where B^i is a rotated version of B^{i-1} . So,

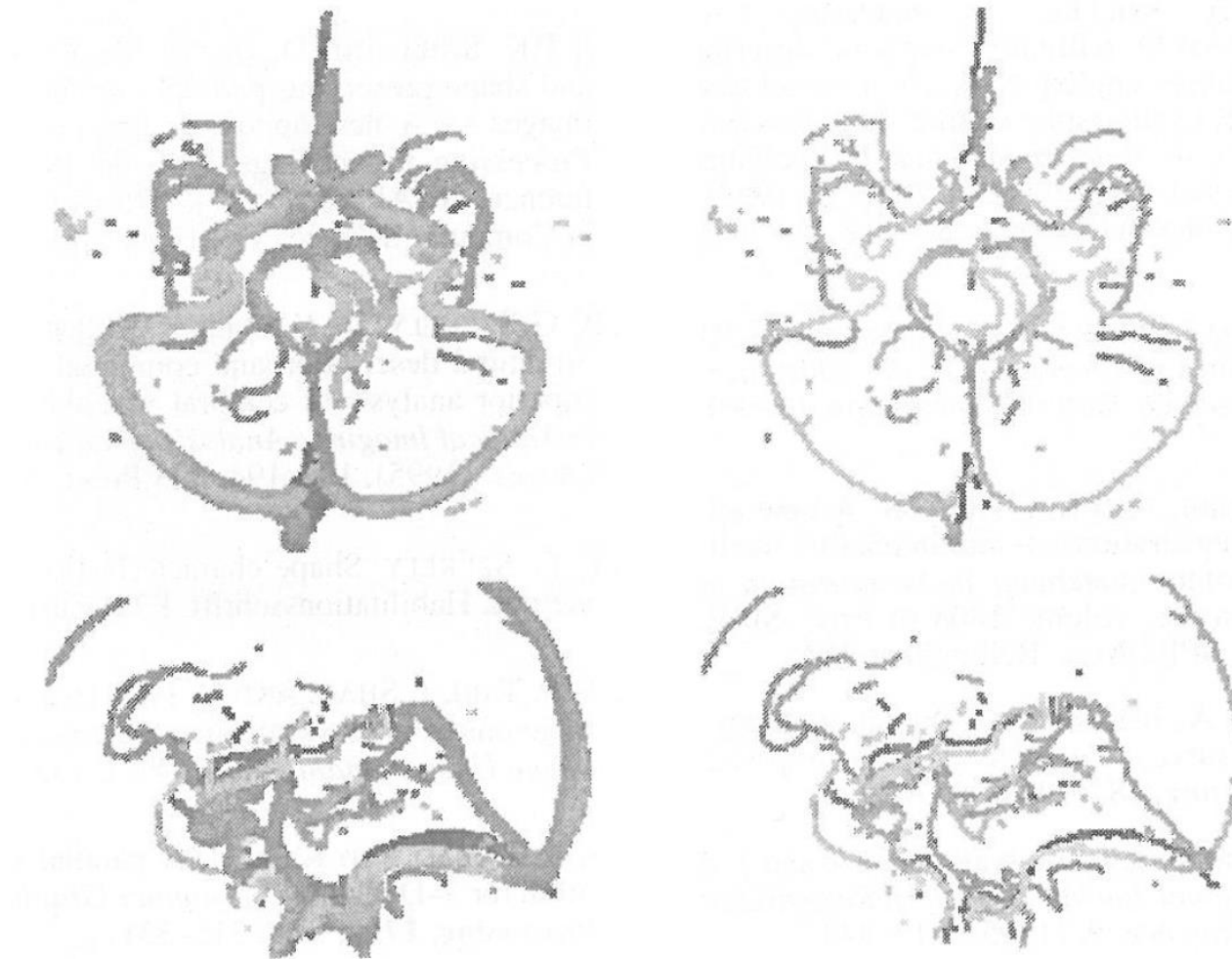
$$A \otimes \{B\} = ((\dots ((A \otimes B^1) \otimes B^2) \dots) \otimes B^n)$$

The entire process is repeated until no further changes occur.

Thinning Process



Thinning Process



Thinning in 3D MRA

Thickening

Thickening is the morphological dual of thinning and is defined by:

$$A \odot B = A \cup (A \circledast B)$$

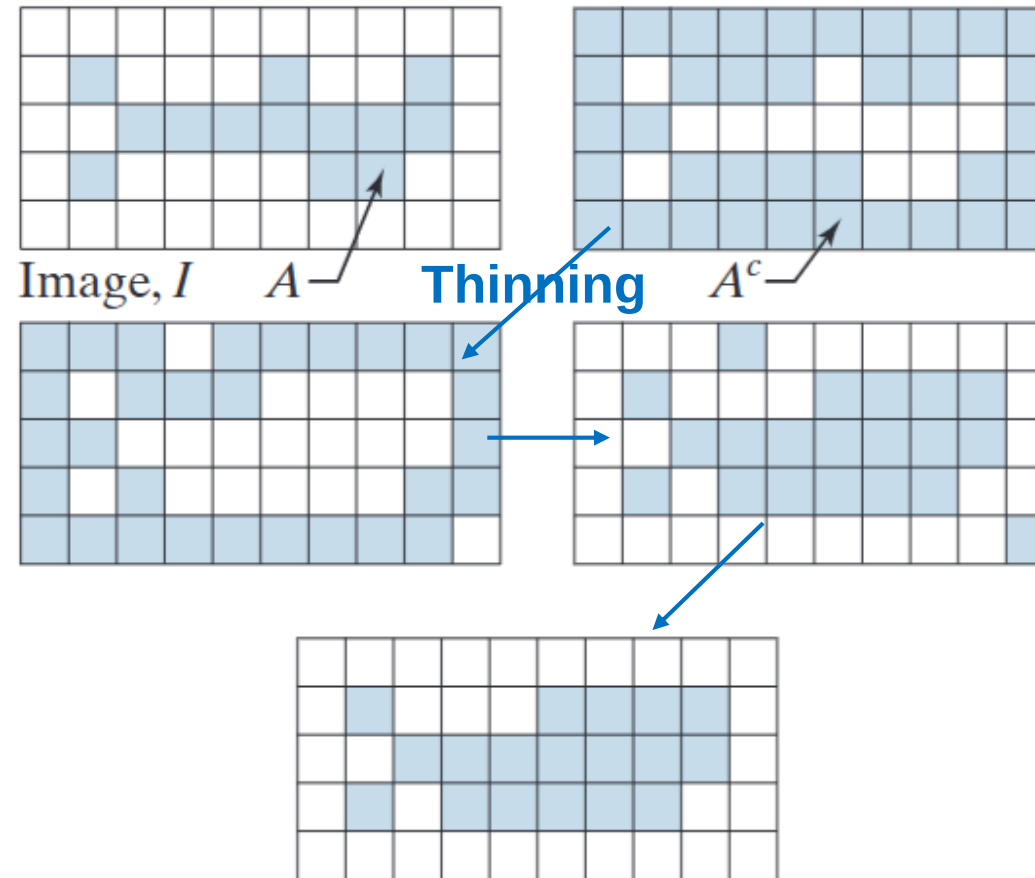
$$\text{Thinning} \quad A \cap (A \circledast B)^c$$

As in thinning, thickening can be defined as a sequential operation

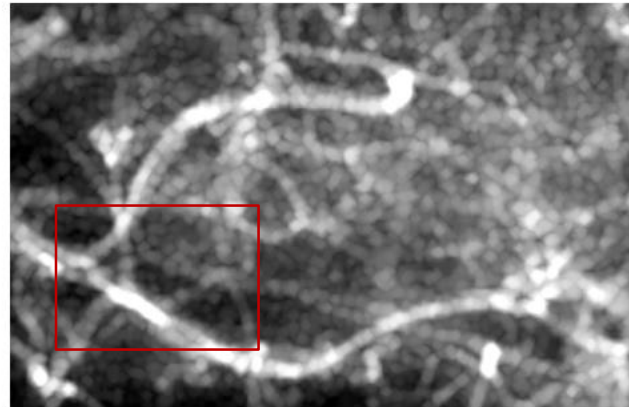
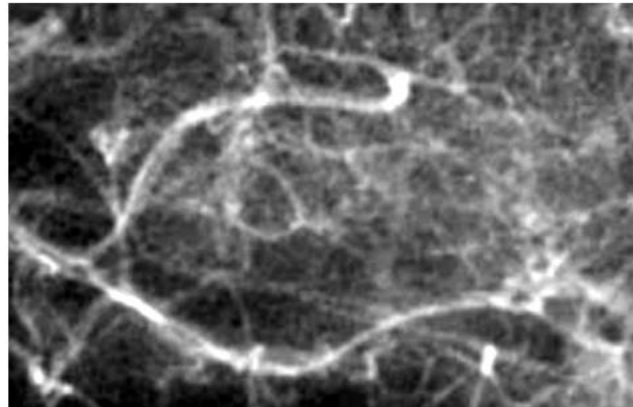
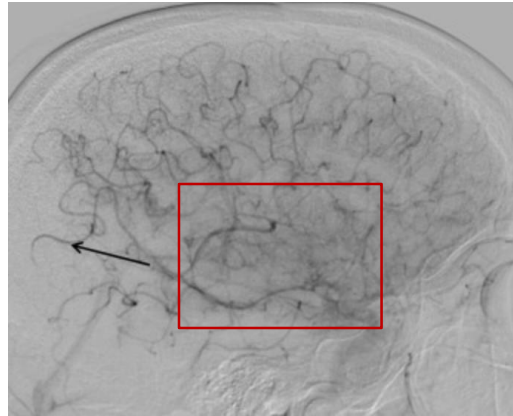
The structure elements used for thickening have the same form as those shown in the thinning example, with all 1's and 0's interchanged. But in practice, we often use the following procedure:

$C = A^c$,
thin C ,
form C^c as the result

Thickening Process



Thickening Process



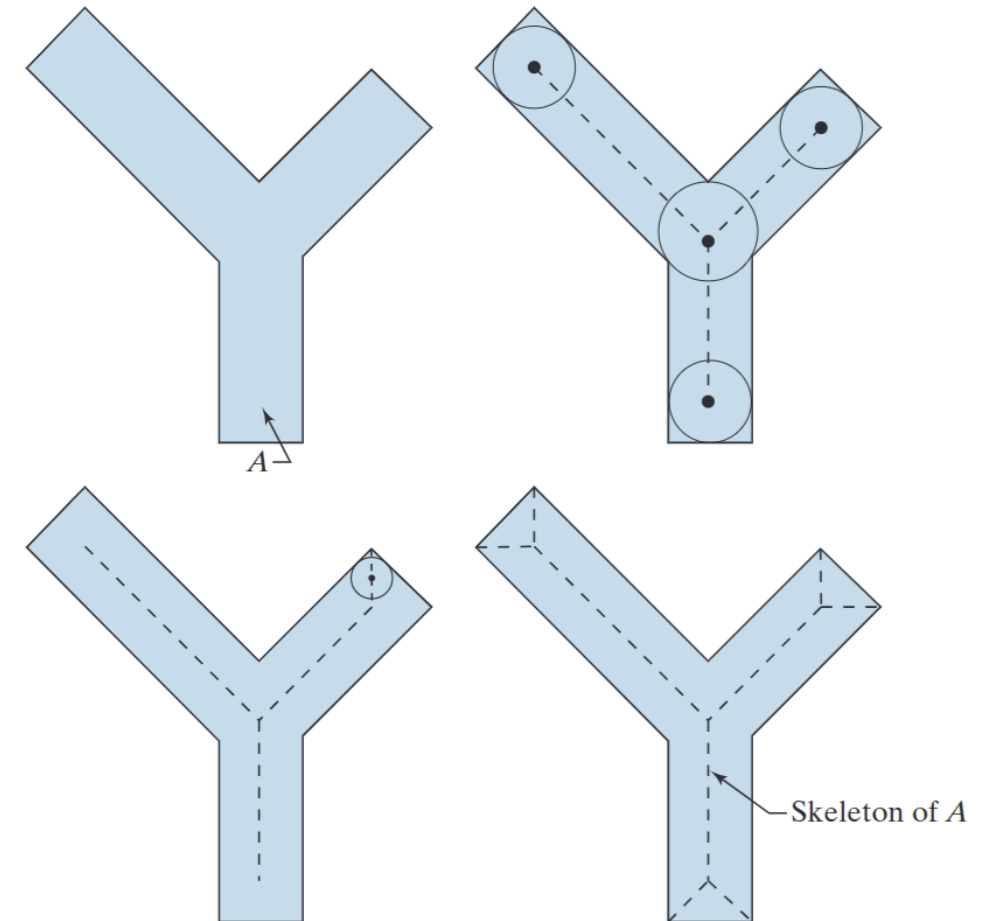
Skeletons

The skeleton of A can be expressed in terms of erosions and openings

$$S(A) = \bigcup_{k=0}^K S_k(A)$$

$$S_k(A) = (A \ominus kB) - (A \ominus kB) \circ B$$

where B is a structure element
 $(A \ominus kB)$ indicates k successive erosions
 starting with A



- If z is a point of $S(A)$, and $(D)_z$ is the largest disk centered at z and contained in A , one cannot find a larger disk (not necessarily centered at z) containing $(D)_z$ and simultaneously included in A . A disk $(D)_z$ satisfying these conditions is called a *maximum disk*.
- If $(D)_z$ is a maximum disk, it touches the boundary of A at two or more different places.

Extensions to Gray-Scale Images

Extending to gray-scale images the basic operations of dilation, erosion, opening, and closing.

$f(x, y)$ is the input image, $b(x, y)$ is a structure element, itself a subimage.

f and b are functions, rather than sets as in the case in binary morphology.

Developing several basic gray-scale morphological algorithms:

- Boundary extraction via a morphological gradient operation
- Region partitioning based on texture content
- Algorithms for smoothing and sharpening

Dilation

Gray-scale dilation of f by b is defined as

$$[f \oplus b_N](x, y) = \max_{(s,t) \in \widehat{b_N}} \{f(x-s, y-t) + \widehat{b_N}(s, t)\}$$

$f(x)$	1	2	2	3	0	3	6	5	2	2	1	0
--------	---	---	---	---	---	---	---	---	---	---	---	---

$b(x)$	1	3	2
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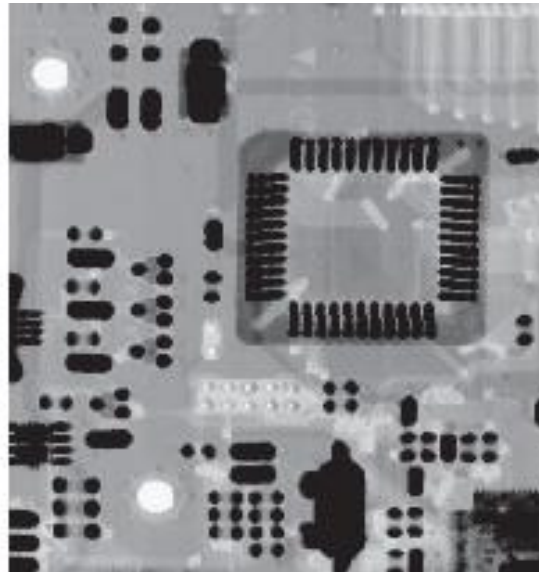
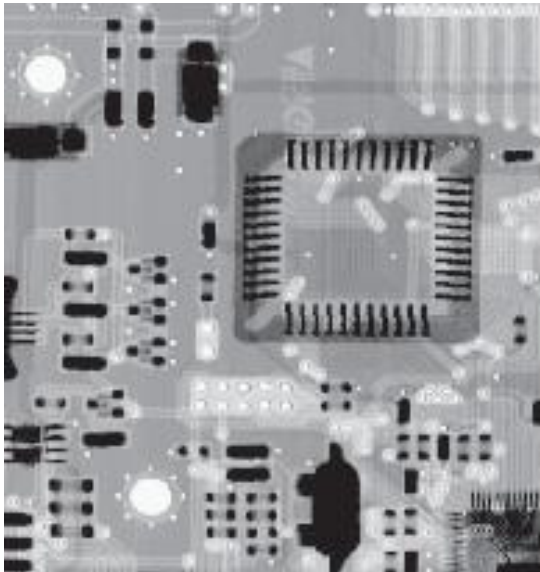
dilation	4	5	5	6	5	8	9	8	6	5	4	3
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Gray-scale erosion of f by b is defined as

$$[f \ominus b_N](x, y) = \min_{(s,t) \in \widehat{b_N}} \{f(x+s, y+t) - \widehat{b_N}(s, t)\}$$

Erosion and Dilation

Erosion

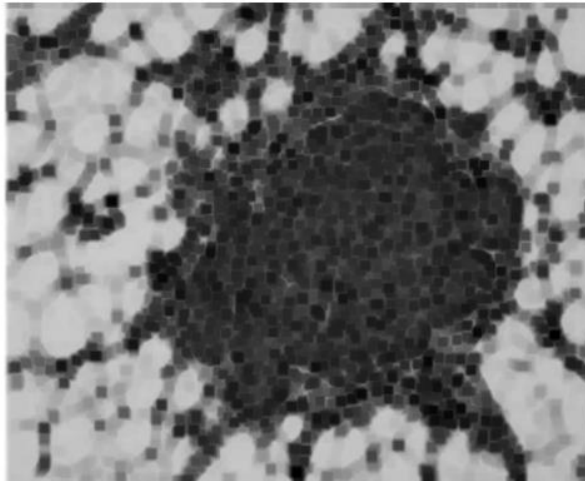
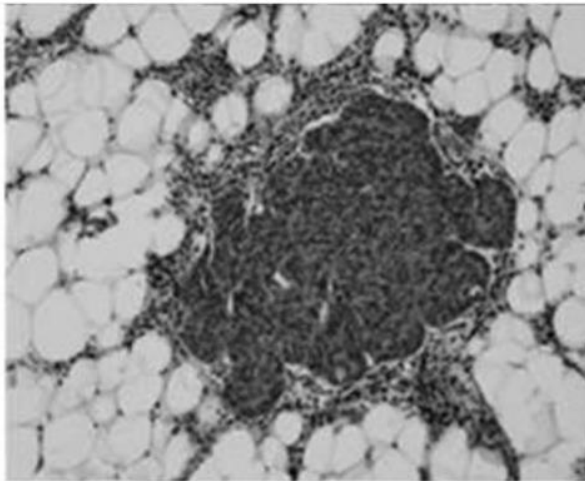


Dilation

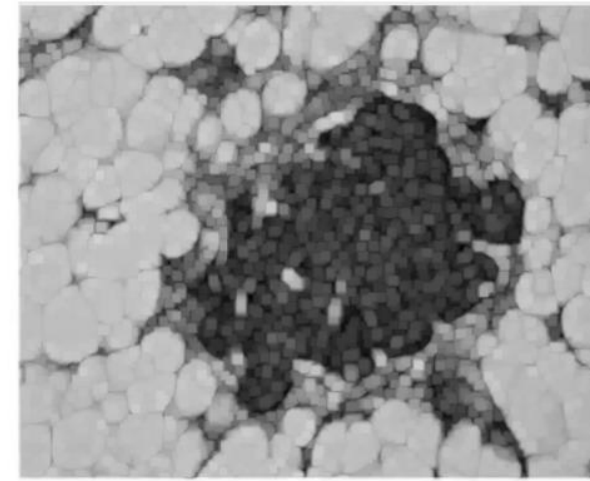


Erosion and Dilation

Erosion



Dilation



Outline

- Binary image
 - Preliminaries
 - Erosion and Dilation
 - Open and close
 - Hit and miss
 - Basic Algorithm (boundary detection, hole filling, connected component, thinning, thickening, skeleton)
- Grey scale image